

Measurement-Based Energy and Throughput Characterization of 28 Embedded Radio Modules and Hardware Variants

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Abstract

This work compares 28 embedded radio modules and physical implementations using a common measurement pipeline and a Nordic Power Profiler Kit II (PPK2). The campaign contains 1644 packet-level operating points, 8220 packet repetitions, and 270 60-second traffic windows for 23 of the entries. It spans sub-GHz and 2.4 GHz radios, direct-SPI transceivers, transparent UART modems, integrated radio microcontrollers, LoRa, 2-(G)FSK, OOK, SimpleLink Long Range, and IEEE 802.15.4g-derived profiles. TX event and RX window energies are reported for every logical payload size exercised by each campaign: 8, 16, 32, 58, 60, 64, 128, 512, or 1024 bytes. The common 32-byte point remains useful for a cross-module comparison, but it no longer substitutes for the measured payload curves. Matched sub-studies compare three E07 (CC1101) implementations, the E07 family against two earlier CC1101 boards, E79 (CC1352P) measurements made through ESP32 and CH9340C UART paths, Ai-Thinker RA-08 (ASR6601) against Ai-Thinker RA-09 (STM32WLE5), CC1101 revisions and bands, E32 bands, SX1278 board implementations, Ai-Thinker RA-02 (SX1278) capacitor variants, NRF24L01 (nRF24L01+) PA/LNA integration, and seven CC1352P PHY profiles. At the common 32-byte point, measured transmit energy spans 0.0187 mJ to 160.4 mJ. Larger-payload results expose fragmentation and integration-window limits; historical E79 CH9340C fragmented-TX values remain window-only observations pending all-frame TX measurements. Continuous delivery results are reported as end-to-end system behavior and are not treated as standalone RF sensitivity measurements.

1 Introduction

Selecting a radio for a battery-powered embedded system is not a one-dimensional choice. Nominal transmit power, data rate, receiver current, modulation, host interface, packet framing, buffering, and firmware latency all influence the energy needed to move an application payload. Datasheet currents are indispensable for design, but they do not include every module-level regulator, level shifter, external PA/LNA, transparent-UART queue, or firmware decision. Conversely, a bench measurement without a controlled workload can make two radios look comparable when their payload, airtime, or reliability differs.

This study treats the measured module as the system boundary and asks six questions:

1. How widely do packet TX/RX energy and sustained-traffic power vary across the tested modules?
2. How do logical payload size and configured rate trade against packet energy and delivered throughput?
3. How much variation remains when the radio IC and LoRa parameters are held constant but the physical implementation changes?
4. What changes when a single programmable radio, the E79/CC1352P, is moved among GFSK, OOK, SimpleLink Long Range (SLR), and IEEE 802.15.4g profiles?
5. How do the E07 CC1101 modules compare with one another and with the two earlier CC1101 boards under common packet settings?
6. How do Ai-Thinker RA-08 (ASR6601) and Ai-Thinker RA-09 (STM32WLE5) compare under matched LoRa settings?

The contribution is an artifact-backed comparison rather than a catalog ranking. Every plotted point is generated from the consolidated CSV files in `power_profiler/comparisons`; the normalized data, plotting script, and LaTeX tables are included beside this manuscript.

2 Devices and Tested Configuration Space

The set includes two earlier CC1101 boards and three Ebyte E07 (CC1101) implementations; direct-SPI and transparent-UART SX127x/SX126x/SX1280 implementations; NRF24L01 (nRF24L01+) modules with and without PA/LNA; an HC-12 (Si4463); the Ebyte E79-400DM2005S (CC1352P) programmable modem under two host-interface contexts; and the Ai-Thinker RA-08 (ASR6601) and Ai-Thinker RA-09 (STM32WLE5) LoRa modems. LoRa-capable chips frequently support additional modulations that were not exercised in their corresponding campaign. Therefore, Table 1 lists the tested PHY, not every mode advertised by the silicon vendor. For example, the SX1276 family can also implement FSK, GFSK, MSK, GMSK, and OOK, while the SX1262 family supports LoRa, (G)FSK, and LR-FHSS [5, 6]. The SX1280 additionally supports FLRC and (G)FSK [7].

Table 1: Measured module and hardware-variant catalog. Rates and powers are the settings exercised in this campaign, not the full capabilities of each chipset.

Module/variant	Radio IC	Band	I/F	Tested PHY/modulation	Tested rates	Tested power
CC1101 V1 433 MHz	CC1101	433 MHz	SPI	2-FSK	1.2/38.4/250 kbps	-30/0/10 dBm
CC1101 V2 868 MHz	CC1101	868 MHz	SPI	2-FSK	1.2/38.4/250 kbps	-30/0/10 dBm
Ebyte E07-400M10S	CC1101	410–450 MHz	SPI	GFSK	1.2/38.4/250 kbps	-30/0/10 dBm
Ebyte E07-433M20S	CC1101 + PA/LNA	425–450.5 MHz	SPI	GFSK	1.2/38.4/250 kbps	-30/0/10 dBm drive
Ebyte E07-900MM10S	CC1101	855–925 MHz	SPI	GFSK	1.2/38.4/250 kbps	-30/0/10 dBm
Ebyte E28	SX1280	2.4 GHz	SPI	LoRa, CR 4/6	SF5/8/12, BW 812.5 kHz	-18/0/13 dBm
Ebyte E280-2G4T12S	SX1280	2.4 GHz	UART	Vendor transparent PHY	1/100/2000 kbps presets	4/7/12 dBm
Ebyte E22-400M30S	SX1268 + PA	433 MHz	SPI	LoRa, CR 4/5	SF7/9/12, BW 125 kHz	-9/10/18 dBm front stage
Ebyte E32-433T20D	SX1278	433 MHz	UART	LoRa transparent, FEC	0.3/4.8/19.2 kbps	10/14/20 dBm
Ebyte E32-433T33D	SX1278 + PA	433 MHz	UART	LoRa transparent, FEC	0.3/4.8/19.2 kbps	24/27/30 dBm
Ebyte E32-868T20D	SX1276	868 MHz	UART	LoRa transparent, FEC	0.3/4.8/19.2 kbps	10/14/20 dBm
Ebyte E32-868T30D	SX1276 + PA	868 MHz	UART	LoRa transparent, FEC	0.3/4.8/19.2 kbps	21/27/30 dBm
Ebyte E79-400DM2005S via ESP32	CC1352P	433 MHz	UART AT via ESP32	2-GFSK, OOK, SLR, 802.15.4g	2.5–200 kbps (7 PHYs)	-20/0/13 dBm
Ebyte E79-400DM2005S via CH9340C	CC1352P	433 MHz	UART AT via CH9340C	2-GFSK, OOK, SLR, 802.15.4g	2.5–200 kbps (7 PHYs)	-20/0/13 dBm
HC-12	Si4463	433 MHz	UART	Transparent (G)FSK	0.5/15/250 kbps presets	-1/8/20 dBm
NRF24L01	nRF24L01+	2.4 GHz	SPI	GFSK, ESB framing	250/1000/2000 kbps	-18/-6/0 dBm
NRF24L01 PA/LNA	nRF24L01+	2.4 GHz	SPI	GFSK, ESB framing	250/1000/2000 kbps	-18/-6/0 dBm drive
Ai-Thinker RA-01H	SX1276	868 MHz	SPI	LoRa, CR 4/5	SF7/9/12, BW 125 kHz	2/10/20 dBm
Ai-Thinker RA-01SH	SX1262	868 MHz	SPI	LoRa, CR 4/5	SF7/9/12, BW 125 kHz	-9/10/22 dBm
Ai-Thinker RA-02	SX1278	433 MHz	SPI	LoRa, CR 4/5	SF7/9/12, BW 125 kHz	-4/10/20 dBm
Ai-Thinker RA-02 with 2 Cap	SX1278	433 MHz	SPI	LoRa, CR 4/5	SF7/9/12, BW 125 kHz	-4/10/20 dBm
Ai-Thinker RA-08	ASR6601	433 MHz	UART AT	LoRa, CR 4/5	SF7/9/12, BW 125 kHz	2/12/22 dBm
Ai-Thinker RA-09	STM32WLE5	433 MHz	UART AT	LoRa, CR 4/5	SF7/9/12, BW 125 kHz	-9/10/22 dBm
Adafruit board with level shifter	SX1278	433 MHz	SPI	LoRa, CR 4/5	SF7/9/12, BW 125 kHz	-4/10/20 dBm
Unshielded radio module	SX1278	433 MHz	SPI	LoRa, CR 4/5	SF7/9/12, BW 125 kHz	-4/10/20 dBm
Carrier PCB with 2 Cap	SX1278	433 MHz	SPI	LoRa, CR 4/5	SF7/9/12, BW 125 kHz	-4/10/20 dBm

Module/variant	Radio IC	Band	I/F	Tested PHY/modulation	Tested rates	Tested power
Shielded radio module	SX1278	433 MHz	SPI	LoRa, CR 4/5	SF7/9/12, BW 125 kHz	-4/10/20 dBm
XL1276-D01	SX1276	433 MHz	SPI	LoRa, CR 4/5	SF7/9/12, BW 125 kHz	-4/10/20 dBm

The E79 campaign is intentionally broader than the others. Its CC1352P supports programmable proprietary formats and multiple modulation families [3]; the measured firmware exposes seven fixed profiles from 2.5 kbps through 200 kbps. The E32 devices, in contrast, expose vendor air-rate presets through a transparent UART interface. Their nominal setting is useful for configuration and pacing, but it is not equivalent to application goodput.

3 Measurement Method

3.1 Instrumentation and boundary

Current was measured with a Nordic PPK2 in ampere-meter mode at a recorded DUT supply of 3.3 V and a sampling rate of 100 kSa.s^{-1} . The PPK2 supports 100 kSa.s^{-1} acquisition and automatically switches current ranges; its documented average-current accuracy is better than $\pm 20\%$ [1]. The latter is an instrument-level bound, not the uncertainty of the sample mean, and should be retained when using small differences for design decisions.

The measured boundary is the powered radio module or radio board under test. Consequently, a result may include the radio IC, module regulator, TCXO, level shifter, onboard MCU, or external PA/LNA, depending on the implementation. USB bridges and the host PC are outside that power boundary. This is deliberate: the goal is deployable module energy, not transistor-level RF-core current. In particular, the E79 ESP32-versus-CH9340C comparison does not measure either bridge’s own supply energy. It tests whether the different UART host path and acquisition session alter the energy observed at the powered E79 boundary.

3.2 Packet campaign

Each packet operating point was repeated five times. Payload matrices were module-appropriate, but all devices include a 32-byte payload. Across the complete dataset, the measured logical payload sizes are 8, 16, 32, 58, 60, 64, 128, 512, and 1024 bytes. The 58-byte E32 and 60-byte HC-12 points reflect campaign-validated frame boundaries; they must not be silently treated as 64-byte observations. The campaign records integrated-window energy, event duration, mean and peak current, sample loss, configuration, status, and packet-delivery telemetry. Larger logical payloads are fragmented when a module has a smaller physical FIFO or validated frame size. The reported energy includes the current within the selected integration window; it includes startup, host delays, or inter-frame gaps only when those intervals lie inside that window. Baseline-subtracted metrics remain available in the source artifacts for analyses focused specifically on incremental activity.

RX window policies differ across acquisition versions. The original CC1101 campaigns enable and disable the receiver within the measured transfer and retain the resulting controlled-RX event window. Later RX campaigns enable listening before capture and integrate a deterministic window whose duration is the modeled airtime. These later values are measured receive/listening-window energies, with a modeled duration; they exclude receiver wake-up and shutdown and do not establish exact RF-synchronized frame boundaries or full host-paced transaction energy. RX values and their recorded delivery outcomes are retained under these definitions.

The E79 ESP32 TX results were reanalysed from all 630 archived current traces. The 315 single-frame observations retain their original windows and values after independent reproduction from RAW. For the 315 fragmented observations, one modeled-airtime window is aligned to each independently detected sustained current burst. Their union preserves the original total integration duration while including every frame and excluding host gaps. Burst count, window geometry, and segmentation sensitivity at three baseline-derived thresholds are checked; no repetitions were discarded. The maximum total-energy sensitivity was 0.002735%, which is a segmentation check rather than a PPK uncertainty estimate. The acquisition code now uses the same method and withholds energy when its frame checks fail.

The E79 CH9340C TX observations at 128, 512, and 1024 bytes retain their historical selected-window energies. Those windows do not establish the energy of all fragments. The 315 repetitions remain available as window observations, while corrected all-frame TX-window energies are pending new measurements because their RAW traces were not saved. Marked values must not be ranked as complete all-frame TX-window energies against reanalysed ESP32 observations. Neither the corrected ESP32 frame-window sum nor the historical CH9340C window should be interpreted as the energy of a complete transaction including host gaps.

The Ebyte E07-400M10S (CC1101), Ebyte E07-433M20S (CC1101 + PA/LNA), Ebyte E07-900MM10S (CC1101), Ebyte E79-400DM2005S via CH9340C (CC1352P), and Ai-Thinker RA-09 (STM32WLE5) additions currently contain packet campaigns only. They are included in the packet tables and matched packet figures, while sustained-power and continuous-delivery fields are shown as not measured. No zero-valued continuous observations are synthesized.

For a current trace $I(t)$ at supply voltage V , event energy over the retained, non-overlapping integration windows $[a_j, b_j]$ is

$$E_{\text{event}} = V \sum_j \int_{a_j}^{b_j} \max(0, I(t)) dt. \quad (1)$$

The clipping follows the acquisition calculation; a deterministic RX or single-frame window has one term. For secondary normalization, the useful-bit metric for payload length L bytes is

$$e_b = \frac{E_{\text{event}}}{8L}. \quad (2)$$

It is retained in the generated CSV for derived analysis, but it is not the principal table metric. A single e_b value can obscure the actual energy of a message and any fragmentation boundary. Therefore, the manuscript reports measured E_{event} in millijoules at each tested payload size. Preamble, sync, headers, coding, CRC, command latency, and fragmentation are intentionally not counted as useful bits in the secondary metric.

3.3 Continuous traffic campaign

Continuous tests use 60s windows with repeated frames and a campaign-defined host gap. Average current and power are computed over the active window,

$$\bar{I} = \frac{1}{T} \int_0^T I(t) dt, \quad \bar{P} = V \bar{I}. \quad (3)$$

The comparative power-versus-RF-setting figures also report the increment above each session's 100 ms pre-trigger baseline,

$$\bar{P}_{\text{excess}} = V \max(0, \bar{I} - \bar{I}_{\text{baseline}}). \quad (4)$$

The baseline state depends on the acquisition version. In the original CC1101 sessions RX is enabled inside the trigger, so the pre-trigger reference has RX off. Later RX sessions enable listening before capture, so their reference is the receiver listening before traffic. These states must not both be interpreted as standby. The clamp is part of the acquisition pipeline. Consequently, a zero excess-power point means that the active-window mean did not exceed that session's baseline; it does not mean that the radio consumed zero power. Total power is independent of the baseline label and remains the battery-budget quantity for the tested continuous workload. Received useful goodput is

$$G_{RX} = \frac{8L_c N_{RX}}{T}, \quad (5)$$

where L_c is the useful content per repeated frame and N_{RX} is the number of frames reported by the receiver. The delivery ratio is $100N_{RX}/N_{TX}$.

These windows are sustained host-driven traffic, not necessarily a radio held continuously in the TX state. Average TX power can therefore be low for short packets separated by the host gap. Likewise, lost frames can result from receiver re-arm latency, UART buffering, serial parsing, or power integrity in addition to RF propagation.

3.4 Normalization and audit rules

For every measured payload size, the packet matrix selects the fastest tested packet mode and highest tested configured power for that module. TX and RX rows are required to use the same mode and power. The common cross-module point is the 32-byte column under the same rule. This compares useful design operating points but not equal range, sensitivity, EIRP, occupied bandwidth, or regulatory duty cycle. A dash in a payload table means *not measured*, never zero energy or an interpolated estimate. LoRa gross rate is estimated as

$$R_b = \frac{BW}{2^{SF}} SF \frac{4}{CR_d}, \quad (6)$$

with $CR_d = 5$ for CR 4/5 and $CR_d = 6$ for the E28 CR 4/6 campaign. Transparent-modem and FSK points use their configured rate labels.

For continuous power, the fastest mode actually present among TX windows is selected and the RX row is matched to the same mode and power. This is often the campaign's representative middle mode (for example

SF9), not the fastest packet mode. Aggregate packet and continuous delivery ratios retain the full configuration matrix.

Modern consolidated files contain explicit attempted and received counters. Older files use the number of runs as the denominator. CC1101 V2 has receiver-side delivery telemetry but no TX-measurement peer telemetry; its packet delivery value is therefore computed from the 90 receiver-side repetitions and is marked *RX-only legacy telemetry*. Missing TX telemetry is not counted as RF loss. Of 8220 measured packet repetitions, 7950 have usable delivery telemetry under these rules.

4 Results

4.1 Cross-module benchmark

Tables 2 and 3 report the measured mean energy of one logical packet for every payload size available in each module campaign. Each populated cell is backed by five packet repetitions at the stated mode and power. The tables deliberately retain different payload limits instead of extrapolating modules beyond the tested workload.

Table 2: Measured TX integration-window energy at each module’s fastest tested mode and highest tested configured power. Payload columns are bytes; a dash means that payload size was not measured. A star marks historical E79 CH9340C fragmented-TX windows: full-transfer energy is pending new measurements.

Module/variant	Mode	Configured power [dBm]	Logical payload size [bytes]									
			8 [mJ]	16 [mJ]	32 [mJ]	58 [mJ]	60 [mJ]	64 [mJ]	128 [mJ]	512 [mJ]	1024 [mJ]	
CC1101 V1 433 MHz (CC1101)	250 kbps	10	0.19	—	0.32	—	—	0.73	1.54	6.39	12.87	
CC1101 V2 868 MHz (CC1101)	250 kbps	10	0.12	—	0.20	—	—	0.48	1.04	4.40	8.87	
Ebyte E07-400M10S (CC1101)	250 kbps	10	0.11	—	0.20	—	—	0.49	—	—	—	
Ebyte E07-433M20S (CC1101 + PA/LNA)	250 kbps	10	0.46	—	0.73	—	—	1.48	—	—	—	
Ebyte E07-900MM10S (CC1101)	250 kbps	10	0.09	—	0.17	—	—	0.42	—	—	—	
Ebyte E28 (SX1280)	SF5	13	0.21	—	0.42	—	—	—	1.31	—	—	
Ebyte E280-2G4T12S (SX1280)	2000 kbps	12	0.02	—	0.04	—	—	0.05	—	—	—	
Ebyte E22-400M30S (SX1268 + PA)	SF7	18	9.25	—	17.74	—	—	—	51.66	—	—	
Ebyte E32-433T20D (SX1278)	19.2 kbps	20	7.91	—	17.69	28.47	—	—	66.32	249.79	498.82	
Ebyte E32-433T33D (SX1278 + PA)	19.2 kbps	30	78.92	—	160.35	241.27	—	—	515.97	1902.12	3820.31	
Ebyte E32-868T20D (SX1276)	19.2 kbps	20	7.91	—	17.66	28.36	—	—	66.05	248.88	498.09	
Ebyte E32-868T30D (SX1276 + PA)	19.2 kbps	30	41.75	—	91.65	146.89	—	—	343.36	1298.43	2590.62	
Ebyte E79-400DM2005S via ESP32 (CC1352P)	GFSK200	13	0.08	—	0.11	—	—	0.15	0.30	1.20	2.41	
Ebyte E79-400DM2005S via CH9340C (CC1352P)	GFSK200	13	0.08	—	0.11	—	—	0.16	0.26*	0.89*	1.72*	
HC-12 (Si4463)	250 kbps	20	3.56	—	5.41	—	9.19	—	—	—	—	
NRF24L01 (nRF24L01+)	2000 kbps	0	0.01	0.01	0.02	—	—	—	—	—	—	
NRF24L01 PA/LNA (nRF24L01+)	2000 kbps	0	0.05	0.07	0.09	—	—	—	—	—	—	
Ai-Thinker RA-01H (SX1276)	SF7	20	16.57	—	30.24	—	—	—	85.00	—	—	
Ai-Thinker RA-01SH (SX1262)	SF7	22	9.90	—	18.03	—	—	—	50.60	—	—	
Ai-Thinker RA-02 (SX1278)	SF7	20	9.14	—	16.67	—	—	—	46.67	—	—	
Ai-Thinker RA-02 with 2 Cap (SX1278)	SF7	20	8.84	—	16.17	—	—	—	45.40	—	—	
Ai-Thinker RA-08 (ASR6601)	SF7	22	16.51	—	32.84	—	—	—	98.05	—	—	
Ai-Thinker RA-09 (STM32WLE5)	SF7	22	16.93	—	33.61	—	—	—	100.14	—	—	
Adafruit board with level shifter (SX1278)	SF7	20	12.08	—	23.27	—	—	—	68.09	—	—	
Unshielded radio module (SX1278)	SF7	20	10.61	—	19.36	—	—	—	54.40	—	—	
Carrier PCB with 2 Cap (SX1278)	SF7	20	10.51	—	19.20	—	—	—	54.06	—	—	
Shielded radio module (SX1278)	SF7	20	11.57	—	22.32	—	—	—	65.11	—	—	
XL1276-D01 (SX1276)	SF7	20	10.71	—	21.35	—	—	—	63.61	—	—	

Table 3: Measured RX integration-window energy at each module’s fastest tested mode and highest tested configured power. Payload columns are bytes; a dash means that payload size was not measured. RX uses the campaign’s receive-window policy; modeled airtime windows exclude receiver wake-up and shutdown.

Module/ variant	Radio IC	Mode	Configured	Logical payload size [bytes]									
			power [dBm]	8 [mJ]	16 [mJ]	32 [mJ]	58 [mJ]	60 [mJ]	64 [mJ]	128 [mJ]	512 [mJ]	1024 [mJ]	
CC1101 V1 433 MHz	CC1101	250 kbps	10	5.90	—	6.18	—	—	6.94	8.65	18.78	32.21	
CC1101 V2 868 MHz	CC1101	250 kbps	10	5.73	—	5.99	—	—	6.78	8.39	18.31	31.69	
Ebyte E07-400M10S	CC1101	250 kbps	10	0.21	—	0.26	—	—	0.52	—	—	—	
Ebyte E07-433M20S	CC1101 + PA/LNA	250 kbps	10	0.22	—	0.27	—	—	0.53	—	—	—	
Ebyte E07-900MM10S	CC1101	250 kbps	10	0.19	—	0.23	—	—	0.47	—	—	—	
Ebyte E28	SX1280	SF5	13	0.05	—	0.10	—	—	—	0.30	—	—	
Ebyte E280-2G4T12S	SX1280	2000 kbps	12	0.59	—	0.59	—	—	0.60	—	—	—	
Ebyte E22-400M30S	SX1268 + PA	SF7	18	0.67	—	1.33	—	—	—	3.98	—	—	
Ebyte E32-433T20D	SX1278	19.2 kbps	20	1.93	—	3.08	4.33	—	—	10.80	38.65	77.06	
Ebyte E32-433T33D	SX1278 + PA	19.2 kbps	30	5.64	—	10.37	15.57	—	—	37.48	136.49	274.38	
Ebyte E32-868T20D	SX1276	19.2 kbps	20	1.92	—	3.08	4.33	—	—	10.82	38.64	77.14	
Ebyte E32-868T30D	SX1276 + PA	19.2 kbps	30	3.00	—	4.88	6.92	—	—	19.46	70.34	141.65	
Ebyte E79-400DM2005S via ESP32	CC1352P	GFSK200	13	0.09	—	0.11	—	—	0.14	0.29	1.14	2.28	
Ebyte E79-400DM2005S via CH9340C	CC1352P	GFSK200	13	0.09	—	0.11	—	—	0.15	0.29	1.16	2.33	
HC-12	Si4463	250 kbps	20	0.34	—	0.36	—	0.40	—	—	—	—	
NRF24L01	nRF24L01+	2000 kbps	0	0.05	0.05	0.06	—	—	—	—	—	—	
NRF24L01 PA/LNA	nRF24L01+	2000 kbps	0	0.09	0.09	0.09	—	—	—	—	—	—	
Ai-Thinker RA-01H	SX1276	SF7	20	1.56	—	2.86	—	—	—	8.04	—	—	
Ai-Thinker RA-01SH	SX1262	SF7	22	0.65	—	1.19	—	—	—	3.32	—	—	
Ai-Thinker RA-02	SX1278	SF7	20	1.76	—	3.24	—	—	—	9.16	—	—	
Ai-Thinker RA-02 with 2 Cap	SX1278	SF7	20	1.76	—	3.24	—	—	—	9.18	—	—	
Ai-Thinker RA-08	ASR6601	SF7	22	1.55	—	3.09	—	—	—	9.22	—	—	
Ai-Thinker RA-09	STM32WLE5	SF7	22	0.73	—	1.46	—	—	—	4.36	—	—	
Adafruit board with level shifter	SX1278	SF7	20	1.45	—	2.92	—	—	—	8.78	—	—	
Unshielded radio module	SX1278	SF7	20	1.37	—	2.73	—	—	—	8.17	—	—	
Carrier PCB with 2 Cap	SX1278	SF7	20	1.40	—	2.79	—	—	—	8.35	—	—	
Shielded radio module	SX1278	SF7	20	1.48	—	2.97	—	—	—	8.94	—	—	
XL1276-D01	SX1276	SF7	20	1.46	—	2.93	—	—	—	8.83	—	—	

At the common 32-byte point, TX energy spans more than four orders of magnitude: 0.0187 mJ for NRF24L01 (nRF24L01+) at 2 Mbps and 0 dBm, versus 160.35 mJ for Ebyte E32-433T33D (SX1278 + PA) at its 19.2 kbps, 30 dBm setting. This is not a same-link-budget comparison. It quantifies the cost of each module's fastest measured, maximum-power operating point for an actual 32-byte application message.

Packet energy across all measured modules

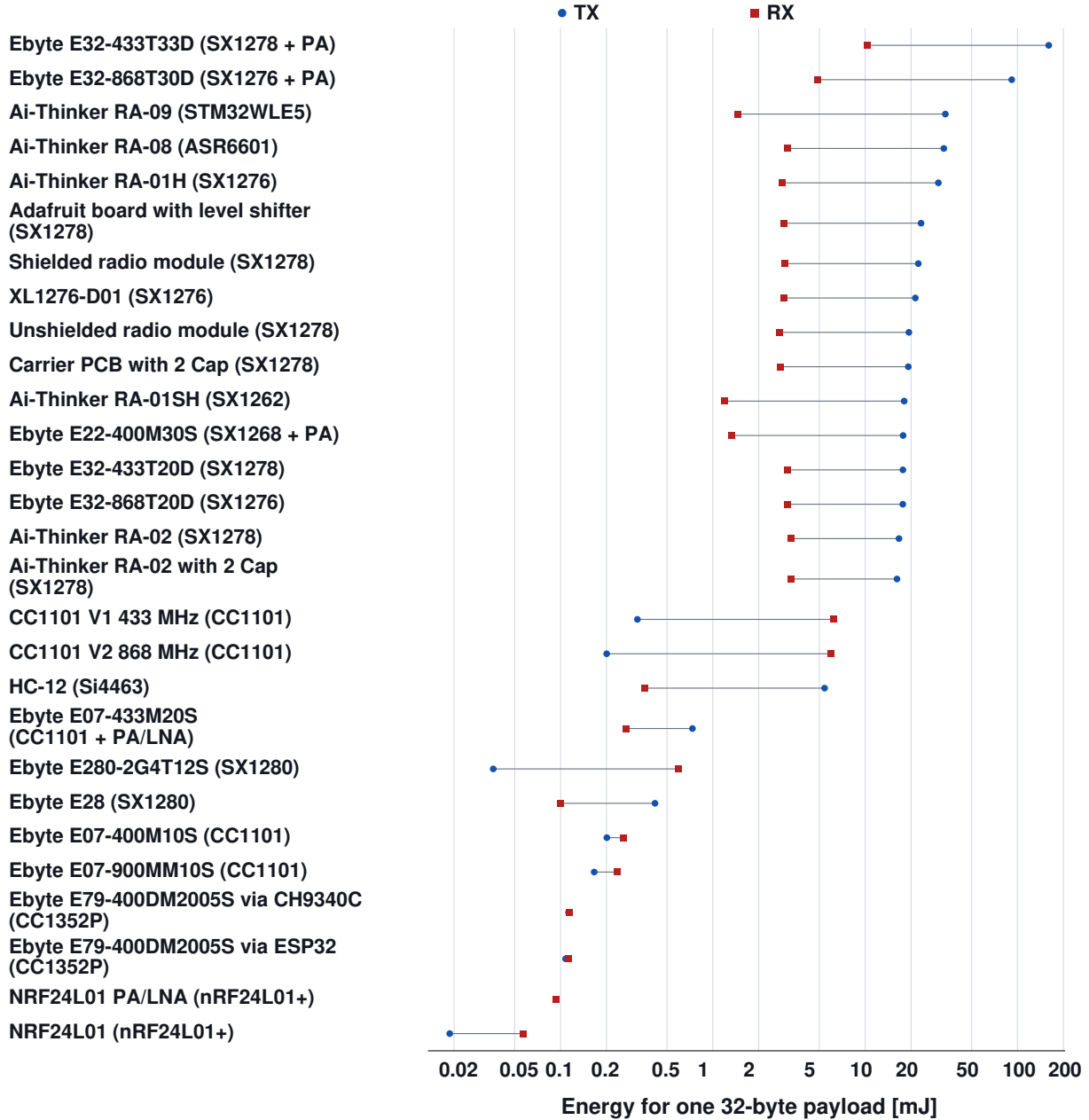


Figure 1: TX event and RX integration-window energy for a common 32-byte application payload. Logarithmic scaling is necessary because both air rate and output stage vary widely.

TX and RX are not symmetric. Fast direct radios such as NRF24L01 (nRF24L01+), Ebyte E28 (SX1280), and Ebyte E79-400DM2005S (CC1352P) have sub-millijoule packet points, while receive integration windows make CC1101 RX energy much larger than its TX event at the fastest rate. The transparent E32 modules show the opposite pattern at high output power: their long, high-current TX path dominates. These differences caution against estimating bidirectional node lifetime from a TX-current headline alone.

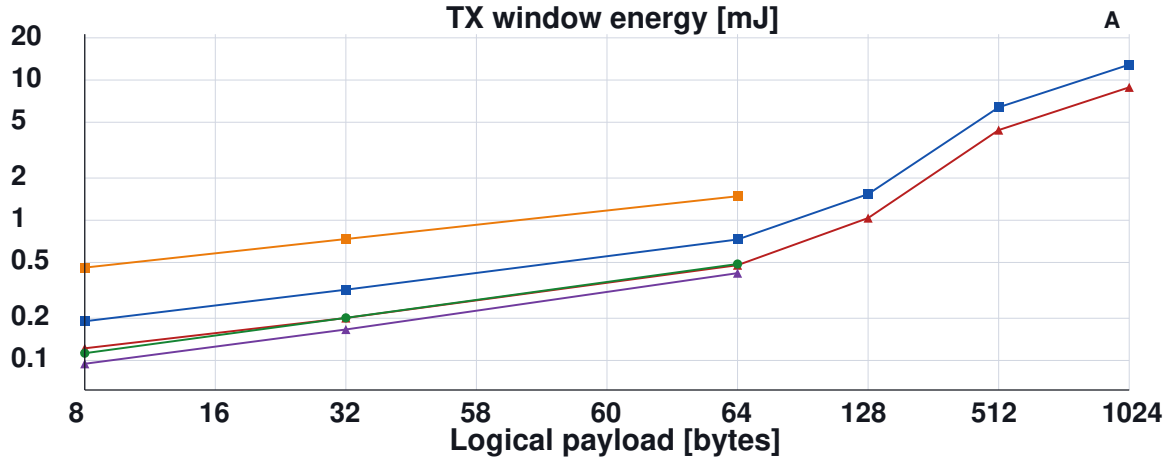
4.2 Payload scaling and fragmentation

TX energy versus measured logical payload size

Fastest tested mode and highest tested configured power per module; logarithmic energy axis

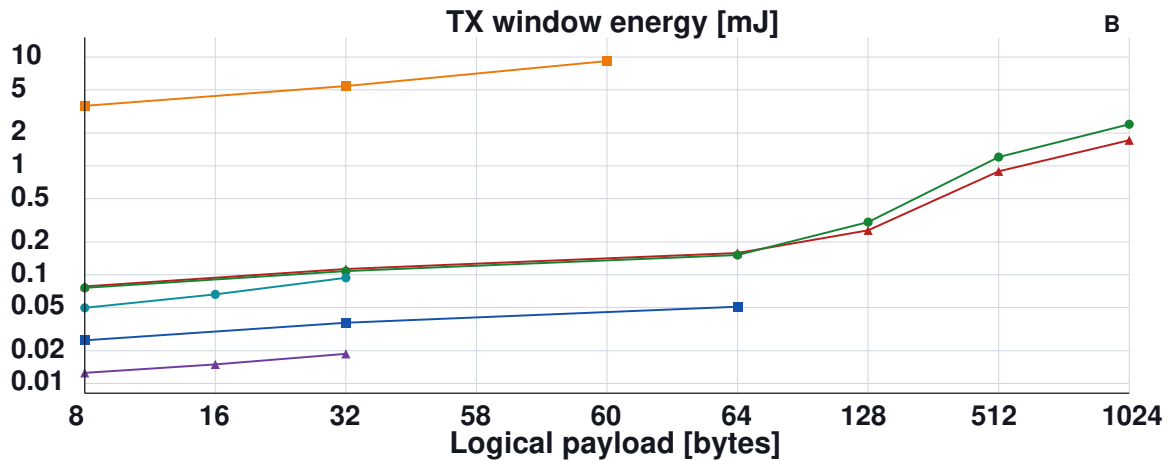
CC1101 boards and E07 modules

- CC1101 V1 433 MHz
- Ebyte E07-400M10S
- ▲ Ebyte E07-900MM10S
- ▲ CC1101 V2 868 MHz
- Ebyte E07-433M20S



FSK and transparent high-rate modules

- Ebyte E280-2G4T12S
- E79 via ESP32
- ▲ NRF24L01
- ▲ E79 via CH9340C *
- HC-12
- NRF24L01 PA/LNA



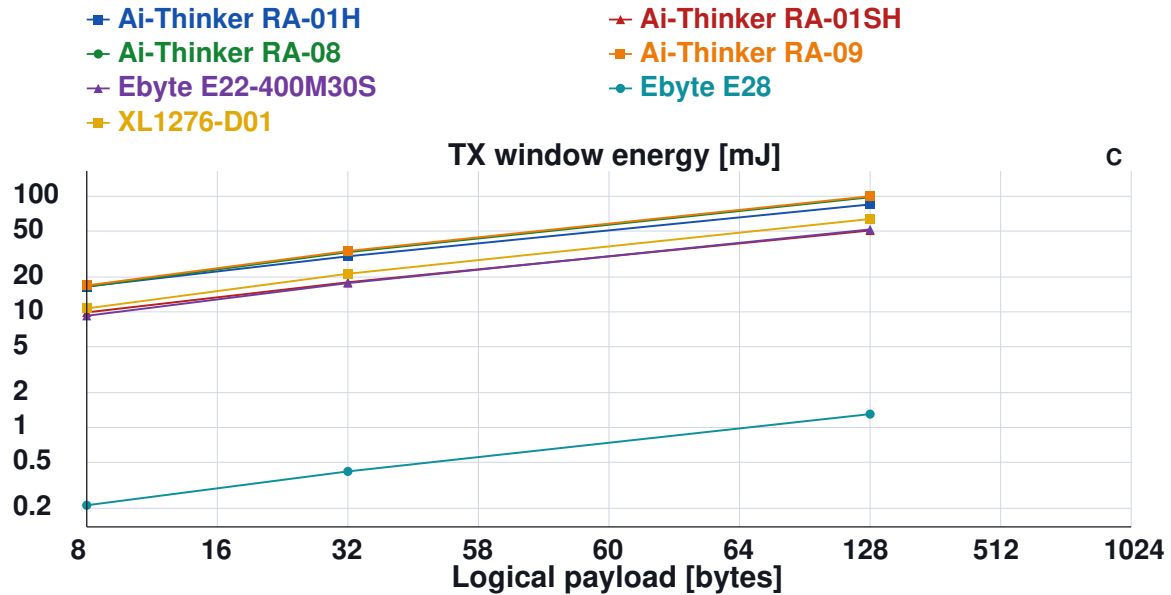
* CH9340C TX above 64 B: historical window only; full-transfer energy awaits new measurements.

Figure 2: Measured TX event energy versus logical payload size, panels A–B. Lines connect measured points only; absent sizes are not interpolated in the numerical tables. Energy uses a logarithmic axis independently in each panel.

TX energy versus measured logical payload size (continued)

Fastest tested mode and highest tested configured power per module; logarithmic energy axis

Direct LoRa and modem implementations



SX1278 physical variants

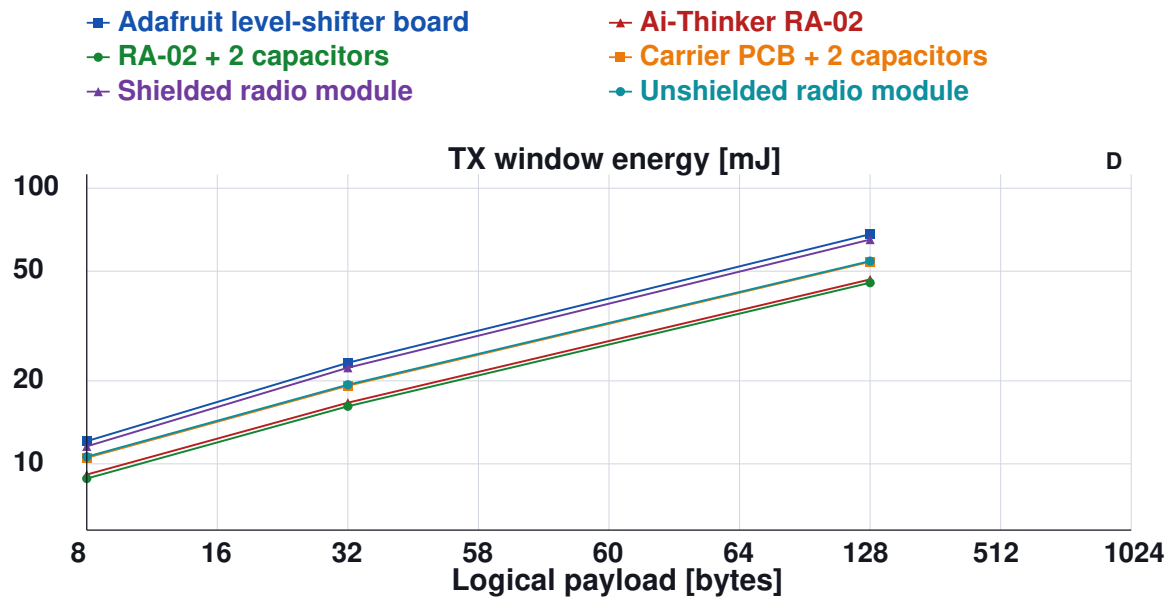


Figure 2: Measured TX event energy versus logical payload size, panels C–D (continued).

TX energy versus measured logical payload size (continued)

Fastest tested mode and highest tested configured power per module; logarithmic energy axis

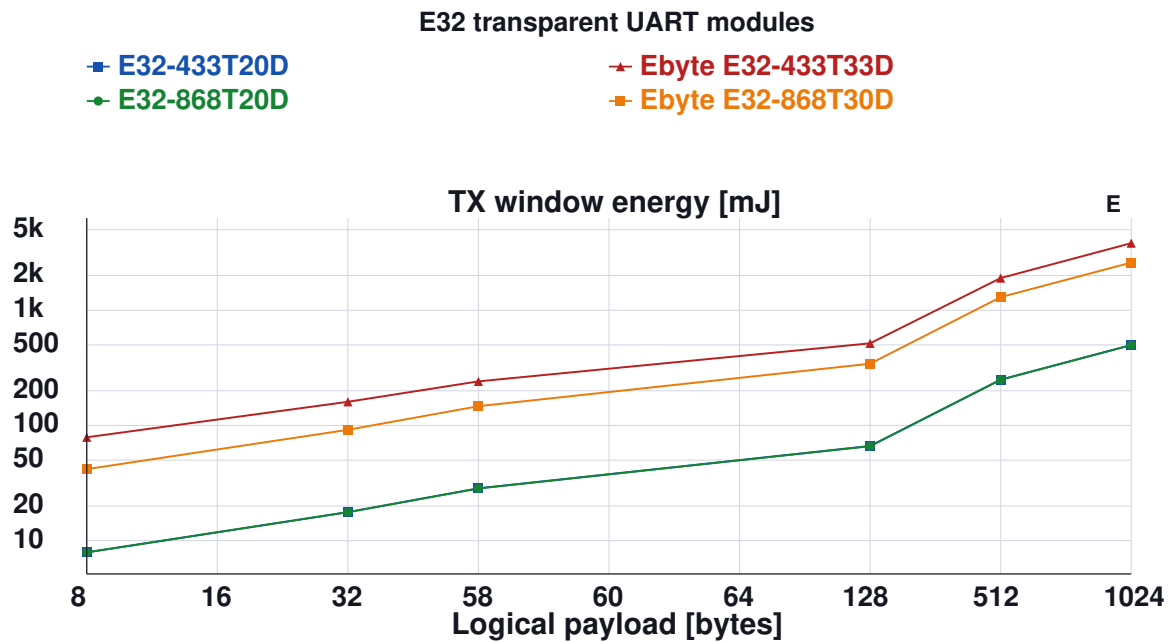


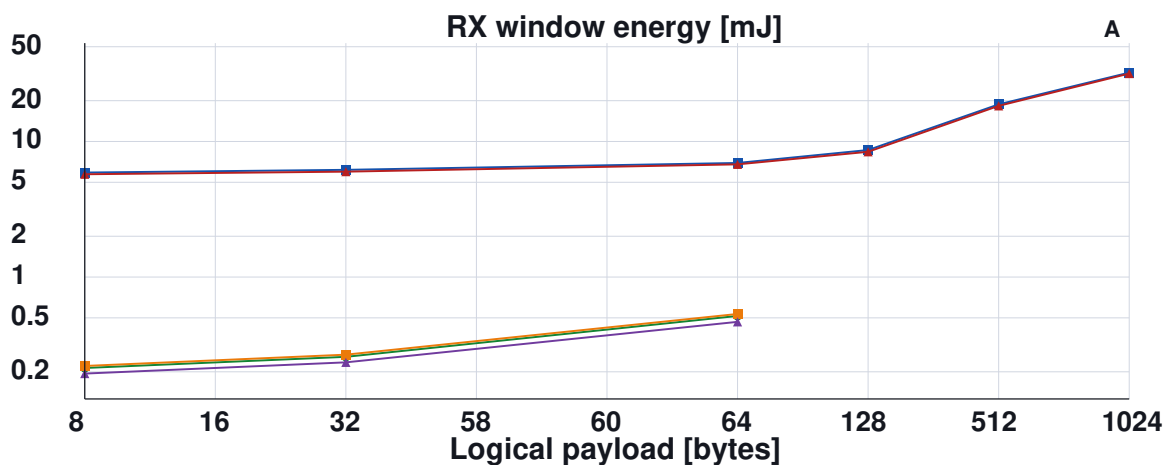
Figure 2: Measured TX event energy versus logical payload size, panel E (continued).

RX energy versus measured logical payload size

Fastest tested mode and highest tested configured power per module; logarithmic energy axis

CC1101 boards and E07 modules

- CC1101 V1 433 MHz
- CC1101 V2 868 MHz
- Ebyte E07-400M10S
- Ebyte E07-433M20S
- ▲ Ebyte E07-900MM10S



FSK and transparent high-rate modules

- Ebyte E280-2G4T12S
- E79 via ESP32
- ▲ NRF24L01
- ▲ E79 via CH9340C
- HC-12
- NRF24L01 PA/LNA

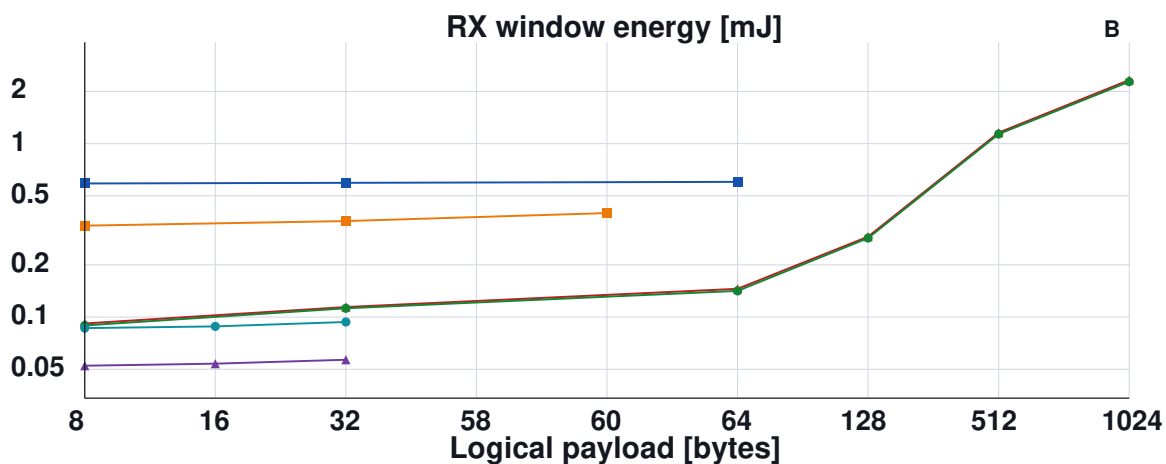


Figure 3: Measured RX integration-window energy versus logical payload size, panels A–B, under the same per-module mode and configured-power rule as the TX matrix.

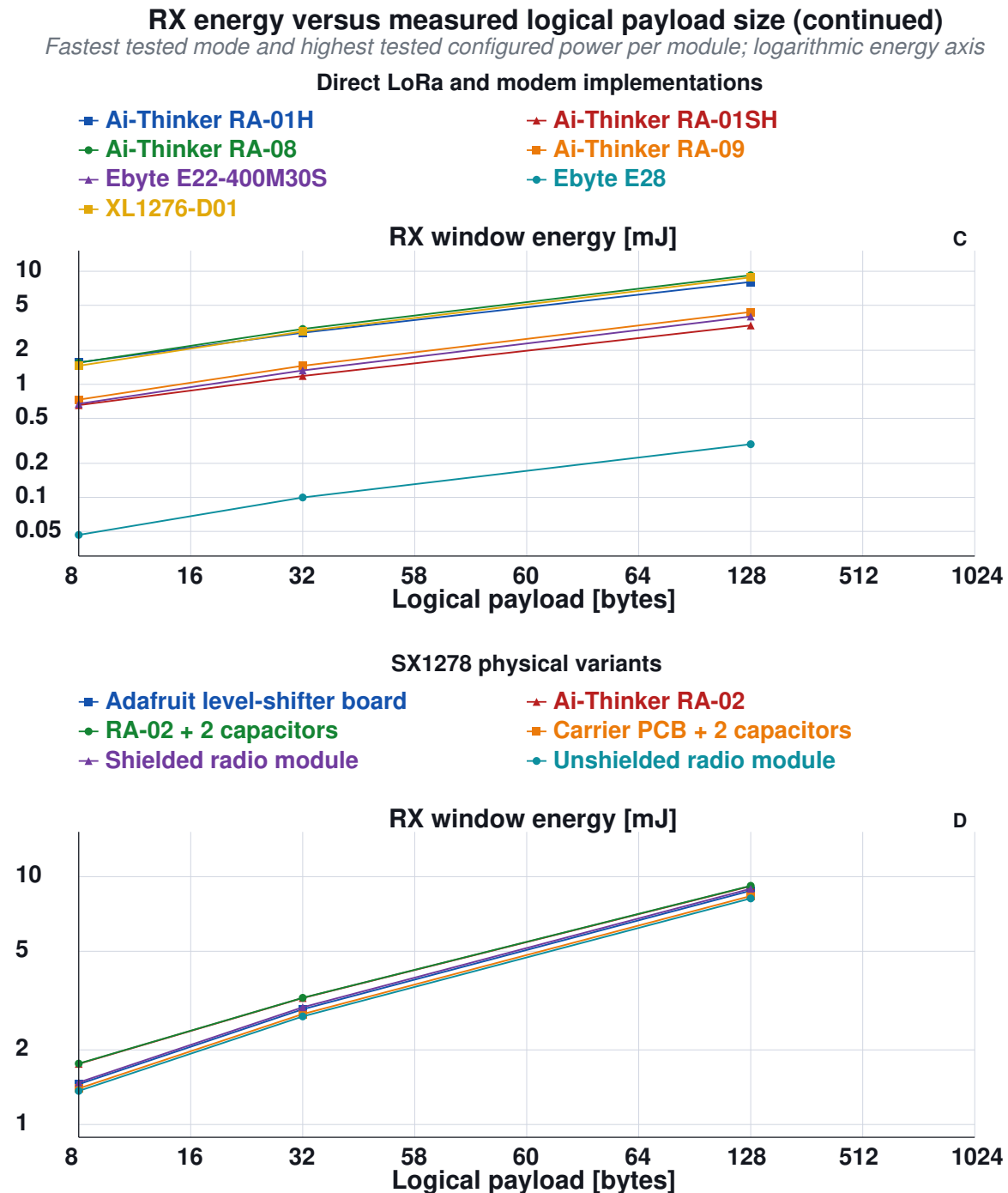


Figure 3: Measured RX integration-window energy versus logical payload size, panels C–D (continued).

RX energy versus measured logical payload size (continued)

Fastest tested mode and highest tested configured power per module; logarithmic energy axis

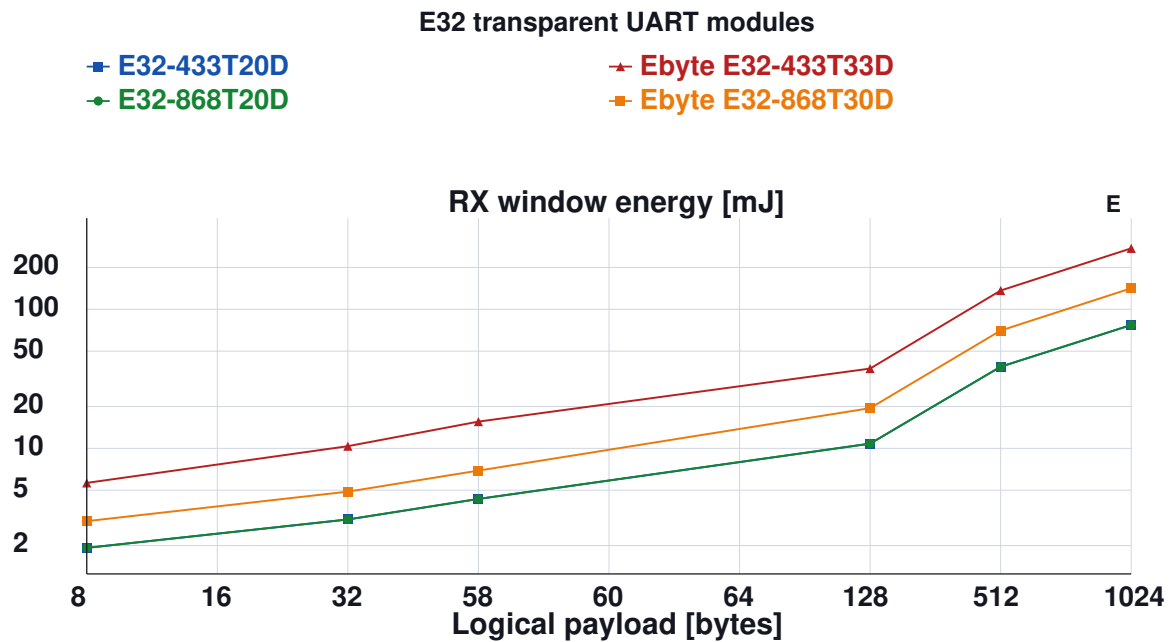


Figure 3: Measured RX integration-window energy versus logical payload size, panel E (continued).

Recorded window energy increases with payload size for every measured TX series. It does not increase in direct proportion to application bytes because the workload, framing, and integration-window policy introduce costs that do not scale directly with payload. Excluding the pending CH9340C fragmented-TX observations, comparison of each module’s smallest and largest measured payload puts the largest-payload TX energy per application byte at 25–57% of its 8-byte value; the corresponding RX range is 4–40%. These ranges describe the reported windows at different payload limits, not equal-size rankings or a guarantee that every window contains a complete transaction. Historical fragmented E79 CH9340C TX points remain explicitly marked pending all-frame TX measurements.

The long-payload campaigns make the practical scale visible. Ebyte E79-400DM2005S via ESP32 (CC1352P), using GFSK200, rises from 0.0757 mJ at 8 bytes to 2.411 mJ at 1024 bytes: a 128-fold payload increase costs 31.9 times more frame-window energy. The corrected 1024-byte value is 45.2% above the historical contiguous-window estimate. Ebyte E32-433T33D (SX1278 + PA) rises from 78.92 mJ to 3820.3 mJ over the same logical sizes. At the opposite extreme, the NRF24L01 (nRF24L01+) hardware payload ends at 32 bytes; its measured TX event rises only from 0.0125 mJ at 8 bytes to 0.0187 mJ at 32 bytes. Application messages larger than the tested native payload limit would require a separate multi-packet experiment and are not extrapolated here.

The 58-, 60-, and 64-byte columns are also intentionally separate. They correspond to E32, HC-12, and other campaign-specific boundaries. The 128-, 512-, and 1024-byte columns identify logical-message sizes that may require multiple over-the-air frames. Their energies retain the integration policy described above: RX remains a defined receive-window metric, and the marked CH9340C TX values are historical partial-window observations. Only measurements covering the complete intended workload support a full-message battery-energy estimate.

4.3 Sustained-traffic power and useful rate

Table 4 separates sustained power, delivered goodput, and aggregate delivery from the packet-size matrices. The configured rate and mode columns belong to the selected continuous window. PDR is the aggregate packet delivery ratio; CDR is the aggregate 60-second continuous delivery ratio. The five packet-only additions retain their measured PDR and use dashes for continuous quantities.

Table 4: Sustained-traffic and delivery benchmark. Continuous values use the fastest mode available in each 60-second campaign. PDR and CDR aggregate the complete packet and continuous matrices, respectively; a dash denotes that no continuous campaign was measured.

Module/variant	Mode	Rate [kbps]	dBm	P_{TX} [mW]	P_{RX} [mW]	Goodput [kbps]	PDR [%]	CDR [%]
CC1101 V1 433 MHz (CC1101)	38.4 kbps	38.40	10.00	69.58	49.77	8.572	92.5	100.0
CC1101 V2 868 MHz (CC1101)	38.4 kbps	38.40	10.00	42.65	46.48	8.892	100.0	90.2
Ebyte E07-400M10S (CC1101)	—	—	—	—	—	—	44.4	—
Ebyte E07-433M20S (CC1101 + PA/LNA)	—	—	—	—	—	—	99.6	—
Ebyte E07-900MM10S (CC1101)	—	—	—	—	—	—	66.7	—
Ebyte E28 (SX1280)	SF8	16.93	13.00	69.31	20.57	9.276	100.0	98.7
Ebyte E280-2G4T12S (SX1280)	100 kbps	100.0	12.00	66.23	62.94	7.374	100.0	40.3
Ebyte E22-400M30S (SX1268 + PA)	SF9	1.758	18.00	191.9	18.15	0.922	93.9	74.8
Ebyte E32-433T20D (SX1278)	4.8 kbps	4.800	20.00	275.9	43.44	1.687	99.7	81.0
Ebyte E32-433T33D (SX1278 + PA)	4.8 kbps	4.800	30.00	252.9	202.7	1.650	83.1	59.5
Ebyte E32-868T20D (SX1276)	4.8 kbps	4.800	20.00	276.4	43.44	1.687	99.7	80.9
Ebyte E32-868T30D (SX1276 + PA)	4.8 kbps	4.800	30.00	1377.0	72.12	1.635	100.0	81.1
Ebyte E79-400DM2005S via ESP32 (CC1352P)	SLR2K5	2.500	13.00	29.34	22.75	2.057	99.9	100.0
Ebyte E79-400DM2005S via CH9340C (CC1352P)	—	—	—	—	—	—	99.8	—
HC-12 (Si4463)	15 kbps	15.00	20.00	102.9	56.39	3.047	100.0	100.0
NRF24L01 (nRF24L01+)	1000 kbps	1000.0	0	1.695	41.80	11.07	78.9	86.5
NRF24L01 PA/LNA (nRF24L01+)	1000 kbps	1000.0	0	8.980	73.30	7.987	64.4	62.2
Ai-Thinker RA-01H (SX1276)	SF9	1.758	20.00	367.4	35.87	1.149	98.3	99.3
Ai-Thinker RA-01SH (SX1262)	SF9	1.758	22.00	217.1	14.76	1.186	100.0	100.0
Ai-Thinker RA-02 (SX1278)	SF9	1.758	20.00	203.7	40.96	1.186	99.4	99.6
Ai-Thinker RA-02 with 2 Cap (SX1278)	SF9	1.758	20.00	196.9	41.07	1.178	99.4	99.4
Ai-Thinker RA-08 (ASR6601)	SF9	1.758	22.00	415.4	42.40	1.212	100.0	100.0
Ai-Thinker RA-09 (STM32WLE5)	—	—	—	—	—	—	100.0	—
Adafruit board with level shifter (SX1278)	SF9	1.758	20.00	255.3	40.63	0.922	88.9	75.4
Unshielded radio module (SX1278)	SF9	1.758	20.00	240.6	37.75	1.169	99.4	70.5
Carrier PCB with 2 Cap (SX1278)	SF9	1.758	20.00	237.7	38.58	1.169	85.0	65.9
Shielded radio module (SX1278)	SF9	1.758	20.00	244.1	41.37	0.93	98.9	76.9
XL1276-D01 (SX1276)	SF9	1.758	20.00	282.9	40.78	1.246	100.0	98.9

The minimum selected TX average is 1.695 mW for NRF24L01 (nRF24L01+), while Ebyte E32-433T33D (SX1278 + PA) reaches approximately 2.525 W. The low NRF24L01 (nRF24L01+) value reflects brief transmissions and idle gaps rather than a permanently enabled carrier. On RX, Ai-Thinker RA-01SH (SX1262) is lowest at 14.76 mW; this agrees directionally with the lower active-RX architecture of the SX1262 generation, whose vendor specification lists a substantially lower receiver current than the SX1276 generation [6, 5].

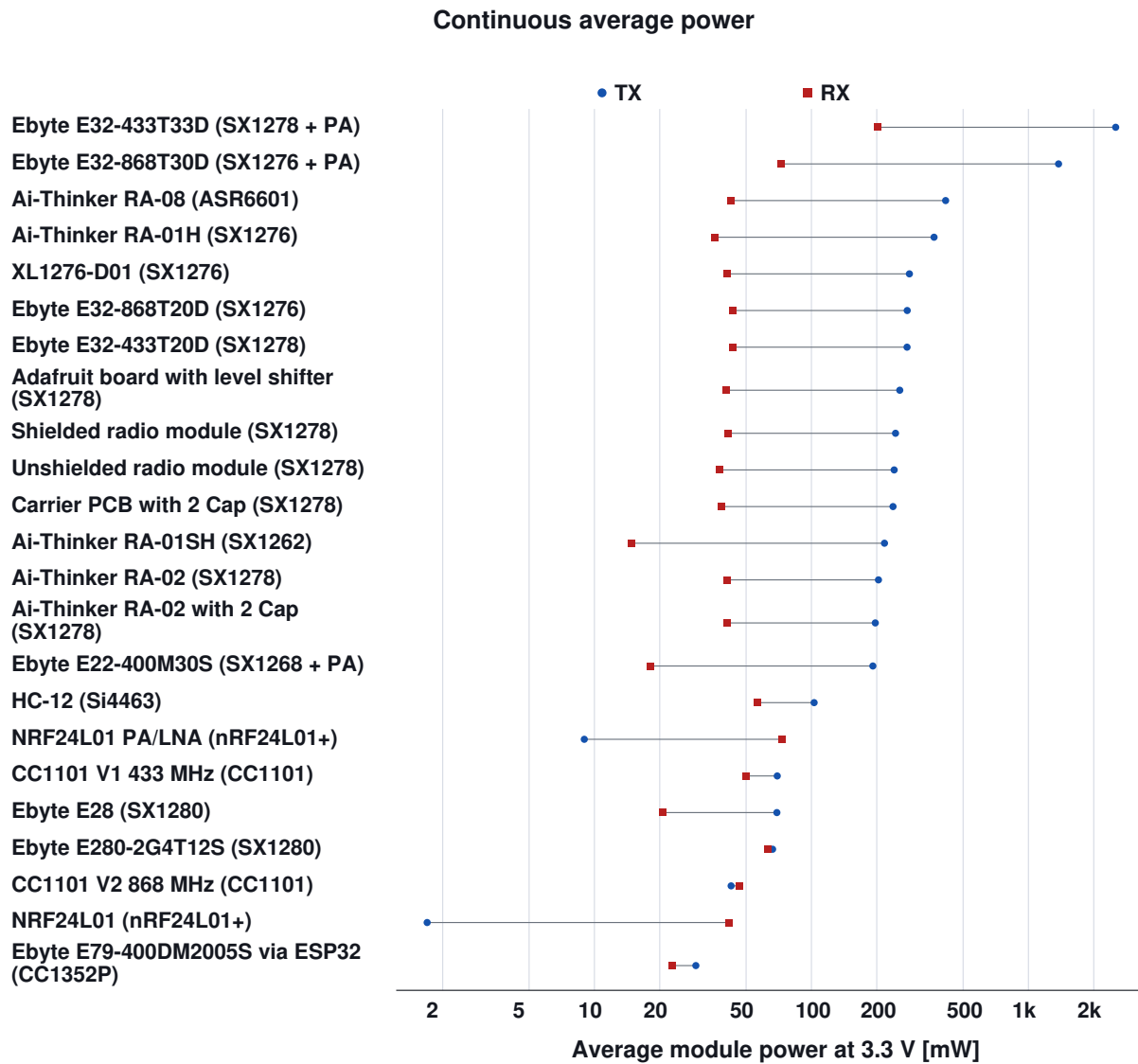


Figure 4: Average module power during the selected 60-second host-driven traffic window. TX and RX use the same mode and configured power within each module.

The highest measured useful goodput in the selected matched windows is 11.07 kbps for NRF24L01 (nRF24L01+). This is far below the radio’s configured 1 Mbps rate because the workload includes a fixed host gap and framing. The result is therefore useful for this bridge and firmware stack, but not a statement about the maximum ESB PHY throughput.

The design space separates into visible regimes. The 2.4 GHz GFSK/transparent devices occupy the high-rate, low-energy region; E79 and CC1101 bridge the middle; LoRa SF7/BW125 modules occupy the lower-rate region; high-power transparent E32 variants pay the largest energy cost. LoRa’s purpose is robust long-range reception rather than minimum energy at a short bench distance, so Figure 5 must not be read as a range-normalized ranking.

4.4 Delivery behavior

Most packet campaigns exceed 98 %, but several hardware/settings combinations are lower: NRF24L01 PA/LNA (nRF24L01+) (64.4 %), NRF24L01 (nRF24L01+) (78.9 %), Ebyte E32-433T33D (SX1278 + PA) (83.1 %), Carrier PCB with 2 Cap (SX1278) (85.0 %), and Adafruit board with level shifter (SX1278) (88.9 %). The campaign audits retain these observations because sample loss was not sufficient to explain them.

Continuous delivery is often lower than packet delivery. The E280 reaches only 40.3 % over its full continuous matrix despite complete packet delivery. Prior controlled checks in the campaign artifacts show that additional receiver/host gap can dramatically reduce loss for slow LoRa configurations. Thus, the continuous metric answers “how did this complete test stack sustain traffic?” rather than “what is the intrinsic RF PER at a specified input level?”

4.5 E79 multi-PHY comparison

The E79 result provides the strongest evidence that airtime is a first-order energy term. A measured 32-byte GFSK200 transmission requires 0.108 mJ, while SLR2K5 requires 4.150 mJ, a 38.5-fold difference on the same hardware, payload, supply, and power setting. At the shared 4.8 kbps nominal rate, OOK4K8 uses 1.763 mJ and GFSK4K8 uses 2.399 mJ. At 50 kbps, GFSK50 and IEEE154G50 are close at 0.270 mJ and 0.283 mJ. Energy alone cannot select among them: SLR trades airtime for coding gain and robustness, while IEEE 802.15.4g and GFSK differ in framing and interoperability.

4.6 E79 host-interface context

The two E79 acquisitions contain observations at a matched TX matrix of 126 points: seven PHY profiles, six payload sizes, and three configured powers. Only the 63 non-fragmented points, at 8, 32, and 64 bytes, are eligible for a like-for-like TX energy comparison without recovering the missing CH9340C fragments. The legacy RX campaign contains 42 points at +13 dBm, all of which have matching CH9340C-context observations; the newer campaign additionally includes RX sweeps at −20 and 0 dBm. Across the paired RX windows, the CH9340C-context to ESP32-context median energy ratio is 1.018. At the plotted 32-byte GFSK200 point, TX changes from 0.108 mJ to 0.113 mJ, while RX changes from 0.112 mJ to 0.114 mJ. The RX modeled window durations are identical within each matched pair, and Figure 8 preserves the PHY ordering. No TX ratio across the entire payload matrix is interpreted as a full-transfer comparison while the CH9340C fragmented measurements remain pending.

The offset among the comparable 32-byte points is useful as a repeatability and integration-context result, but it is not the energy cost of either USB converter. The ESP32 and CH9340C supplies were not inside the measured boundary, and the two campaigns were collected in different sessions. UART edge timing, modem state before the trigger, supply connection, instrument range history, and unit/session variation remain possible causes. Given the PPK2 instrument bound and absence of repeated bridge-swapping trials, the data support “similar with a small session-dependent offset,” not a claim that one USB bridge reduces E79 radio energy.

4.7 E07 family comparison

Figures 9 and 10 isolate the three E07 implementations at a fully delivered common setting. For a 32-byte packet at 38.4 kbps and −30 dBm, Ebyte E07-400M10S (CC1101) and Ebyte E07-900MM10S (CC1101) are nearly identical in TX at 0.376 mJ and 0.379 mJ; their RX energies are 0.708 mJ and 0.653 mJ. Ebyte E07-433M20S (CC1101 + PA/LNA) uses 1.358 mJ TX and 0.736 mJ RX. Its approximately 3.6-fold TX increase relative to the two 10-dBm modules is consistent with measuring the complete module boundary including its external PA/LNA control path, even at a low CC1101 drive command. It does not establish conducted RF output or PA efficiency.

Payload growth increases total event energy for all three modules, while increasing the configured rate sharply reduces airtime and energy. Ebyte E07-400M10S (CC1101) and Ebyte E07-900MM10S (CC1101) track closely over both controlled axes despite their different RF bands. Ebyte E07-433M20S (CC1101 + PA/LNA) remains

Packet energy versus rate design space

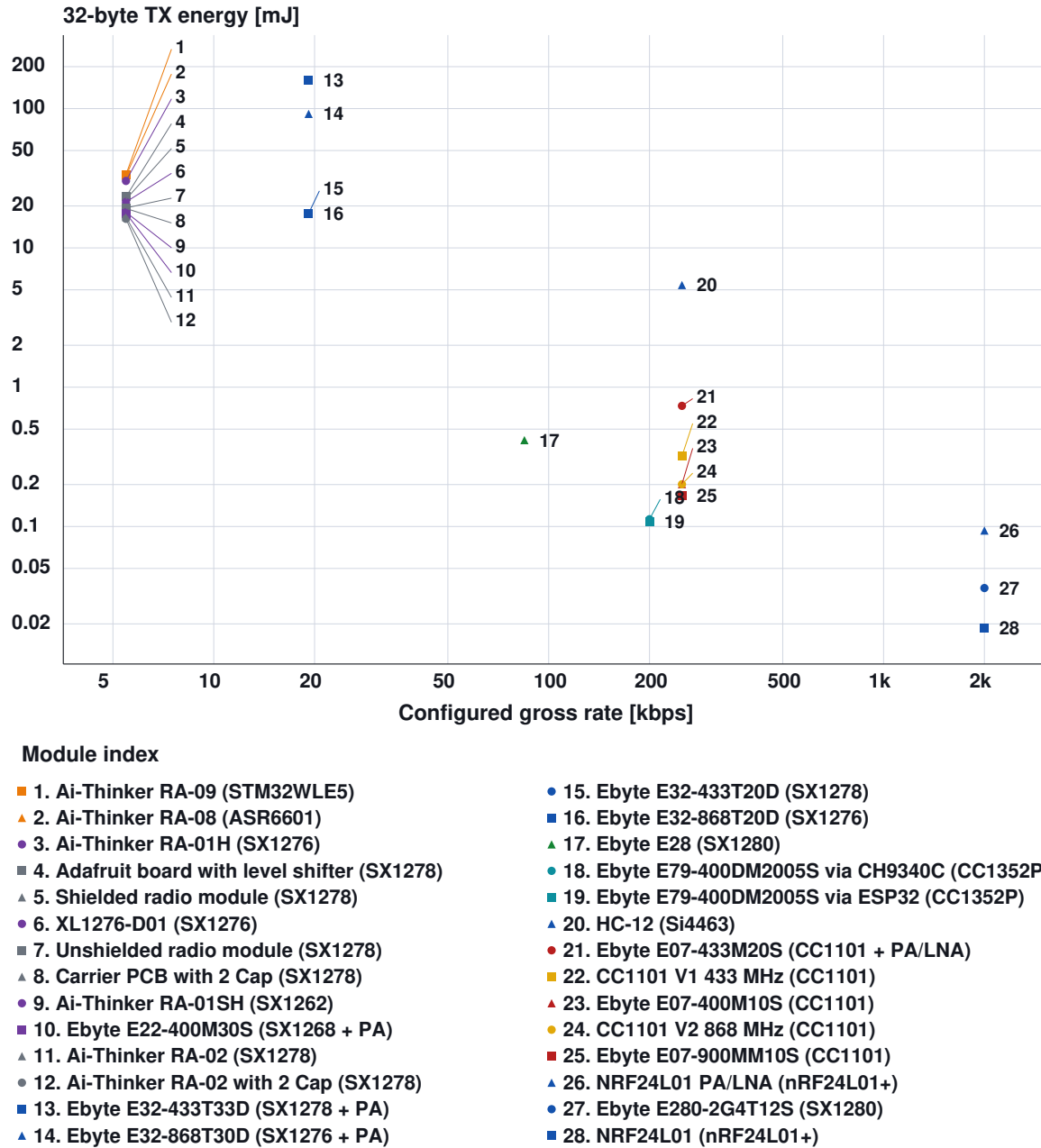


Figure 5: Configured packet-mode rate versus measured TX energy for one 32-byte application packet. Numbered point identifiers correspond to the module index below the plot. Points farther right and lower are preferable for this short-range workload, but the plot does not encode sensitivity, range, interference tolerance, or regulatory limits.

Delivery ratio across campaigns

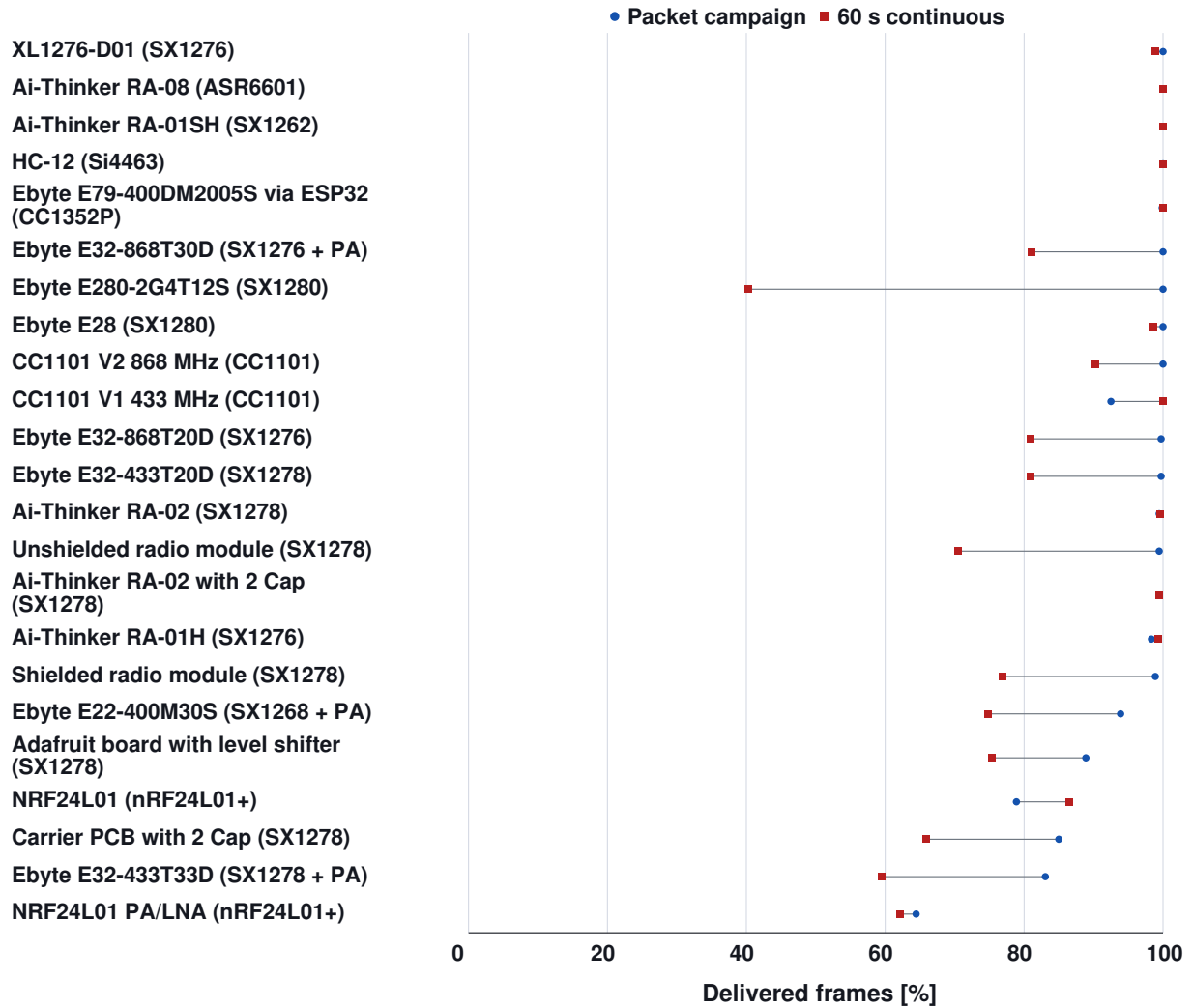


Figure 6: Aggregate packet and continuous delivery ratios for the 23 entries with both campaign types. The continuous ratio is an end-to-end bridge/modem/receiver result and is not a packet-error-rate sensitivity test.

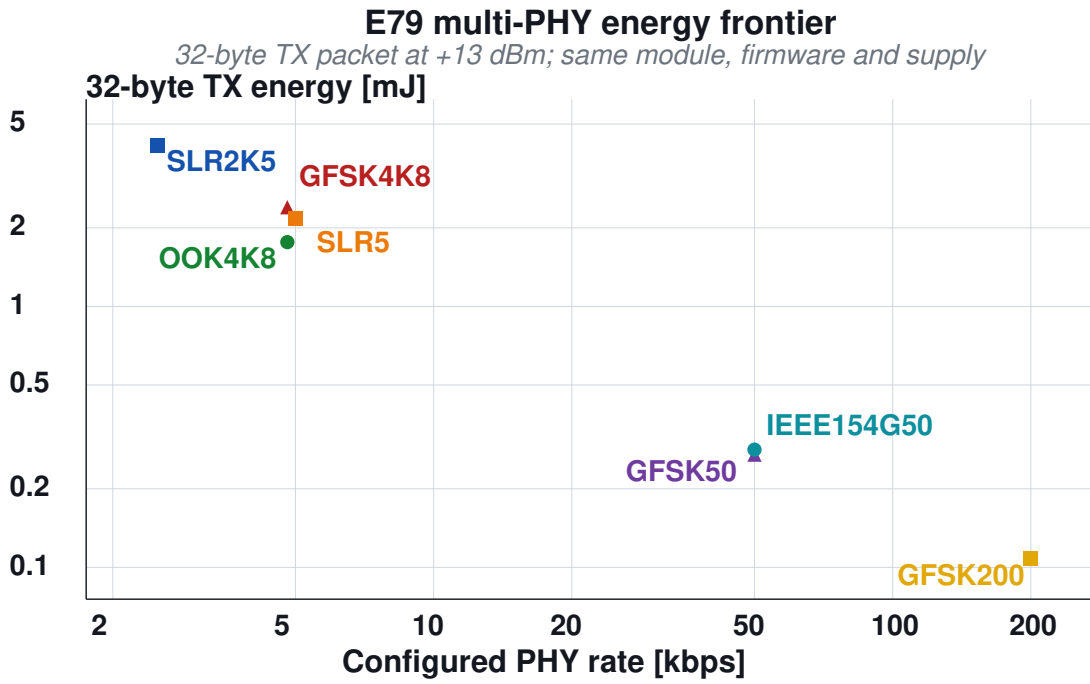


Figure 7: Seven E79/CC1352P PHY profiles on the same module at +13 dBm. This is the cleanest within-device isolation of PHY rate and modulation in the campaign.

E79 interface-context comparison

Same CC1352P matrix point: 32-byte packet, +13 dBm; converter power is outside the PPK2 boundary

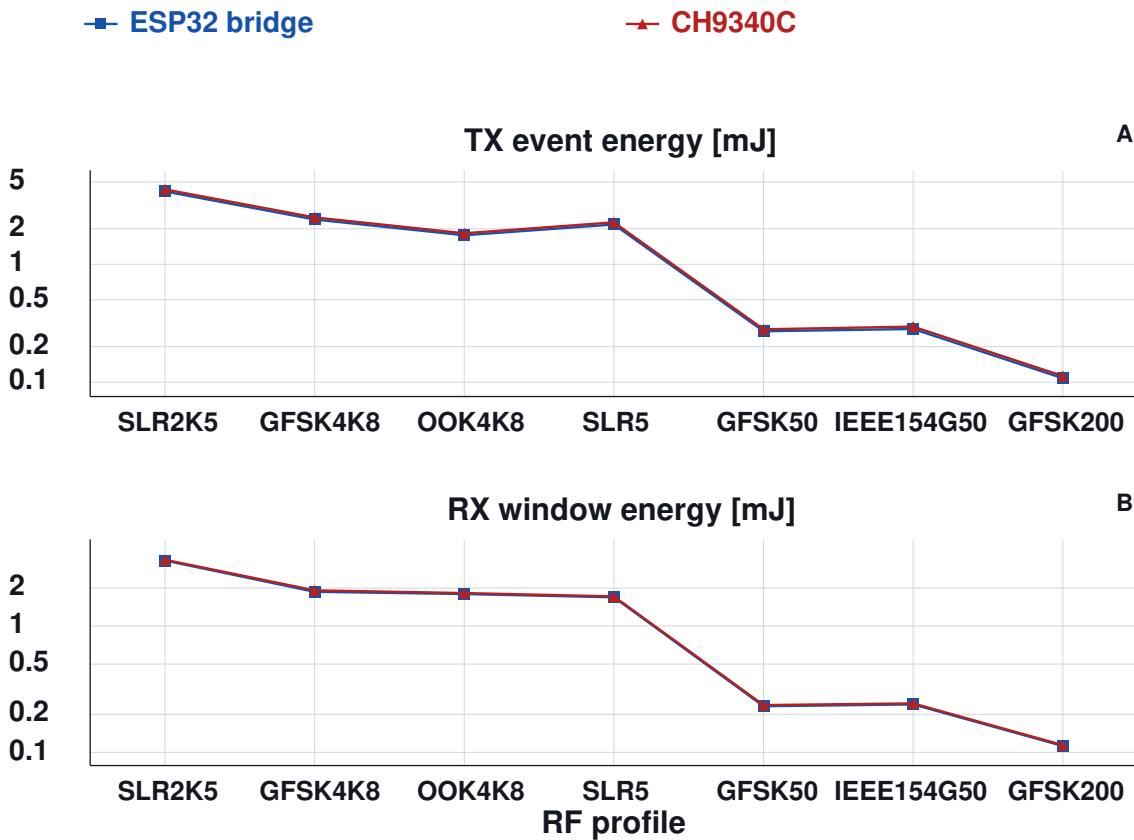


Figure 8: E79 packet energy measured through the original ESP32 UART bridge and the newer CH9340C path. The plotted 32-byte, +13-dBm points use identical PHY settings and modeled event durations. Bridge power is outside the PPK2 boundary.

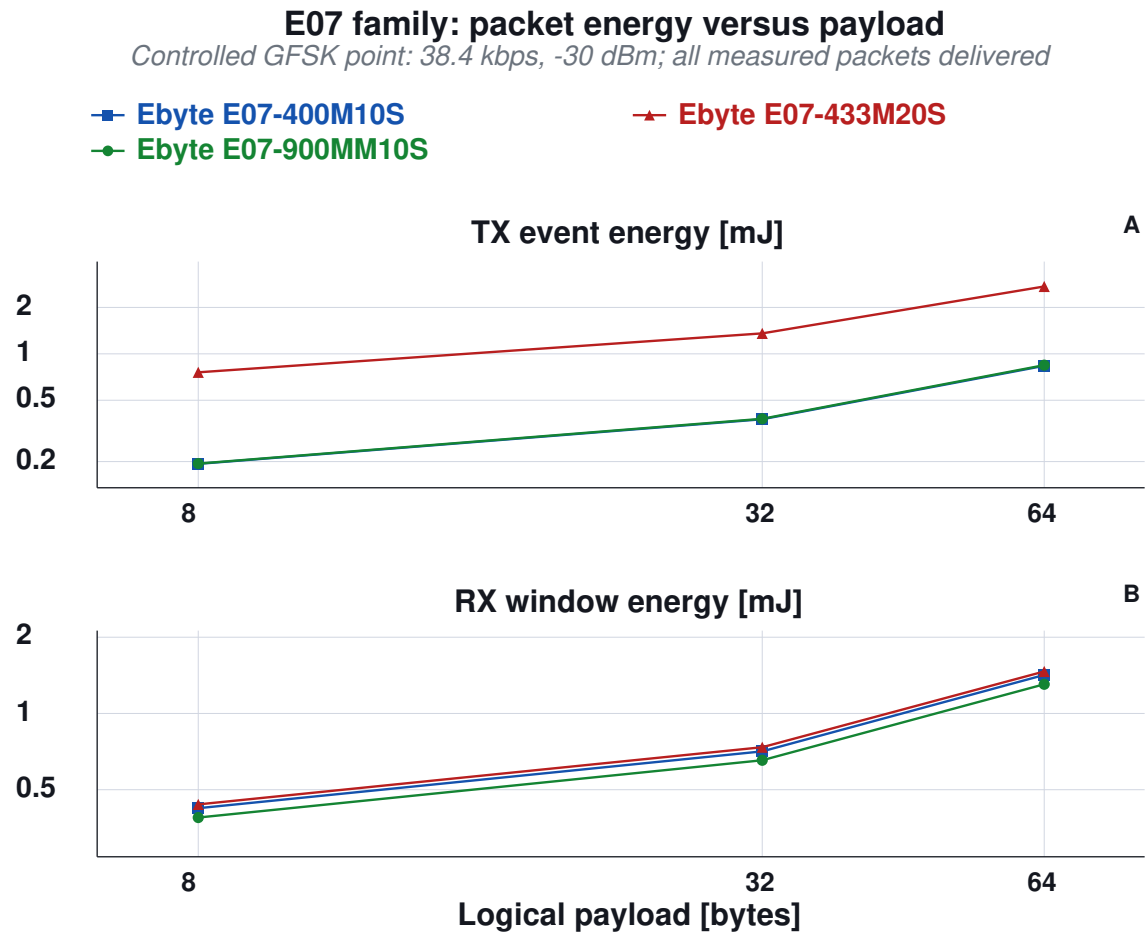


Figure 9: Controlled E07 packet-energy scaling with payload at 38.4 kbps and -30 dBm. Every plotted point delivered all five packets.

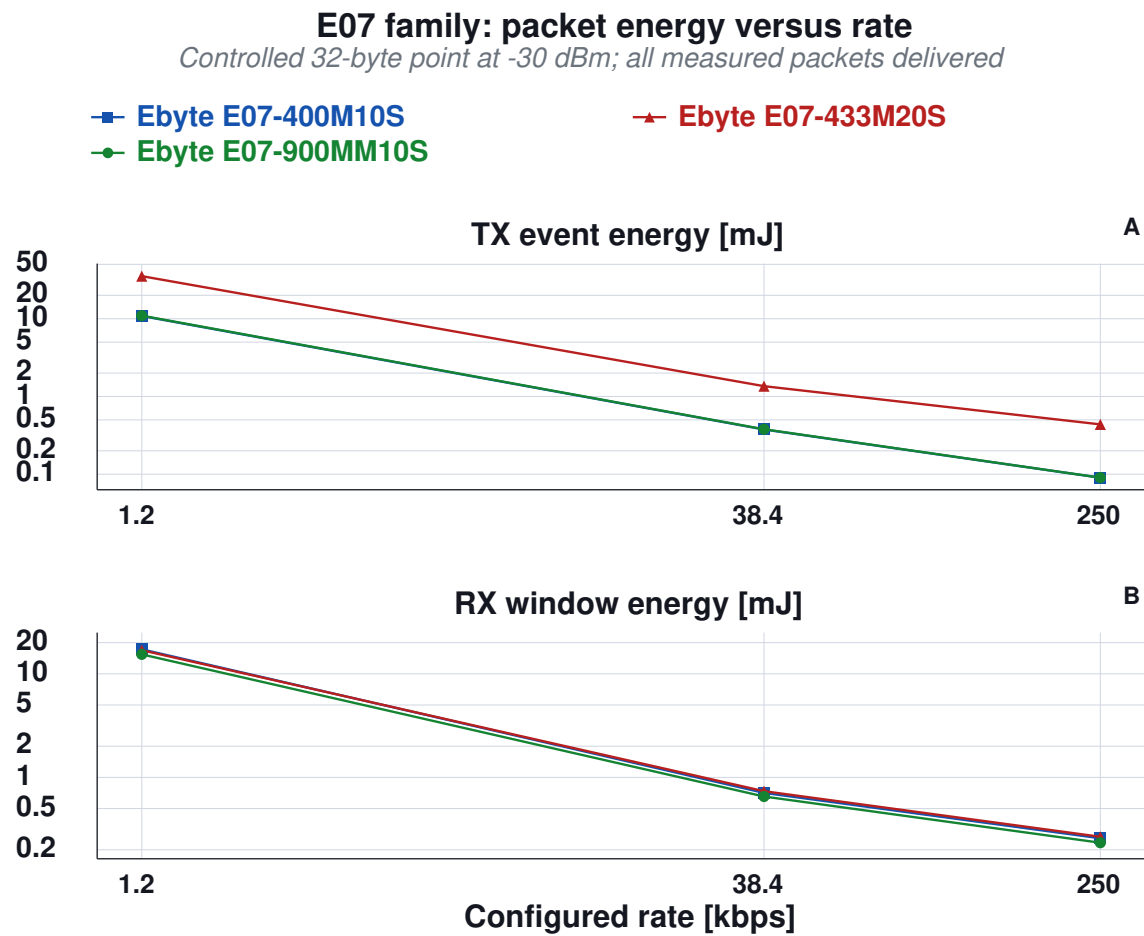


Figure 10: Controlled E07 packet-energy scaling with configured rate for a 32-byte payload at -30 dBm. Every plotted point delivered all five packets.

TX-dominated because its external output stage is part of the powered device, whereas the three RX curves remain much closer.

4.8 CC1101 revision and band comparison

Figures 11 and 12 replace a single-point comparison with the complete common axes of the two campaigns. At 32 bytes, 250 kbps, and +10 dBm, V2 uses 0.201 mJ TX versus 0.320 mJ for V1, a 37.2% reduction. RX energy differs much less: 5.990 mJ versus 6.177 mJ. The pattern persists at 1024 bytes, where V2 uses 8.874 mJ TX versus 12.873 mJ for V1, while RX is 31.695 mJ versus 32.214 mJ. Rate scaling also shows the TX path as the dominant difference: at 1.2 kbps, the 32-byte V2 event is 33.11 mJ, compared with 58.31 mJ for V1.

The continuous result is consistent with the packet result at high configured power. At +10 dBm, V2 averages 42.65 mW TX versus 69.58 mW for V1; after subtracting the session’s pre-trigger baseline, the values are 36.88 mW and 63.70 mW. RX total power remains closer, between 46.3 mW and 47.3 mW for V2 versus 49.8 mW to 51.6 mW for V1. The -30 dBm V2 RX window delivered only 70.7% of transmitted frames, compared with 99.95% for V1, so that point is not equivalent useful work despite its lower mean power.

This is a controlled workload comparison, not a pure PCB-revision experiment: V1 operates at 433 MHz and V2 at 868 MHz. Frequency-dependent matching, configuration, board revision, and unit variation are confounded. The figures establish measured module-level behavior but do not assign its cause.

4.9 E07 modules versus the earlier CC1101 boards

The TX panel provides the strongest cross-campaign comparison because its event durations remain close: approximately 11–13 ms at -30 dBm. At that setting, Ebyte E07-400M10S (CC1101) and Ebyte E07-900MM10S (CC1101) use 0.376 mJ and 0.379 mJ, compared with 0.440 mJ and 0.444 mJ for CC1101 V1 433 MHz (CC1101) and CC1101 V2 868 MHz (CC1101). Ebyte E07-433M20S (CC1101 + PA/LNA) is substantially higher at 1.358 mJ, reflecting its module-level PA/LNA implementation. At +10 dBm the ordering persists, but the Ebyte E07-400M10S (CC1101) and Ebyte E07-900MM10S (CC1101) points do not represent useful delivery in this close-range setup.

The RX panel must be interpreted more cautiously. The E07 event windows are about 12.37 ms, whereas the legacy CC1101 RX windows are about 119 ms; therefore, their absolute RX event energies are not normalized to a common capture duration. The figure retains both because they are real application-level measurements, but it does not attribute the roughly order-of-magnitude energy difference to silicon or PCB efficiency. A future remeasurement with one trigger and receive-window policy would turn this into a controlled five-board RX comparison.

Delivery behavior also constrains the power sweep. Ebyte E07-400M10S (CC1101) lost all packets at the 38.4 kbps 0- and +10-dBm points, while Ebyte E07-900MM10S (CC1101) lost all at +10 dBm; Ebyte E07-433M20S (CC1101 + PA/LNA) delivered this particular matrix slice. The broader Ebyte E07-433M20S campaign contains one lost packet outside the plotted slice. Because low-power -30-dBm points delivered completely, the failures are consistent with close-range overload or setup interaction rather than insufficient link budget, but conducted attenuation is required to establish the cause.

4.10 Ai-Thinker RA-08 (ASR6601) versus Ai-Thinker RA-09 (STM32WLE5)

The two modem campaigns share payloads of 8, 32, and 128 bytes, SF7/SF9/SF12, BW125, CR4/5, and an exact +22-dBm point. Event durations are also matched on RX and differ only slightly on TX, making this one of the stronger cross-chip comparisons in the study. At 32 bytes and SF7, Ai-Thinker RA-09 (STM32WLE5) uses 33.61 mJ TX versus 32.84 mJ for Ai-Thinker RA-08 (ASR6601), a difference of only about 2.4%. Ai-Thinker RA-09 (STM32WLE5) RX is 1.458 mJ versus 3.087 mJ for Ai-Thinker RA-08 (ASR6601), approximately 53% lower over the same 71.93 ms receive window.

The pattern persists across spreading factors. At SF12, TX energy is 829.53 mJ for Ai-Thinker RA-09 (STM32WLE5) and 817.93 mJ for Ai-Thinker RA-08 (ASR6601), while RX is 36.64 mJ and 77.02 mJ, respectively. Thus, the measured distinction is not lower LoRa airtime: the RX integration windows are equal. It is a module-boundary current difference under the tested firmware and supply. Range, sensitivity, and sleep-state transition energy remain outside this comparison.

4.11 Physical implementation effects

Ai-Thinker RA-02 with 2 Cap (SX1278) uses 16.17 mJ versus 16.67 mJ for Ai-Thinker RA-02 (SX1278), a 3.0 % reduction at this point. Unshielded radio module (SX1278) and Carrier PCB with 2 Cap (SX1278) differ by less than 1 % in TX energy. Shielded radio module (SX1278) and Adafruit board with level shifter (SX1278) are higher at 22.32 mJ and 23.27 mJ, respectively, compared with 19.36 mJ for Unshielded radio module (SX1278).

CC1101 V1/V2 continuous-power comparison

60 s windows, 32-byte frames, 38.4 kbps, 15 ms host gap, 3.3 V

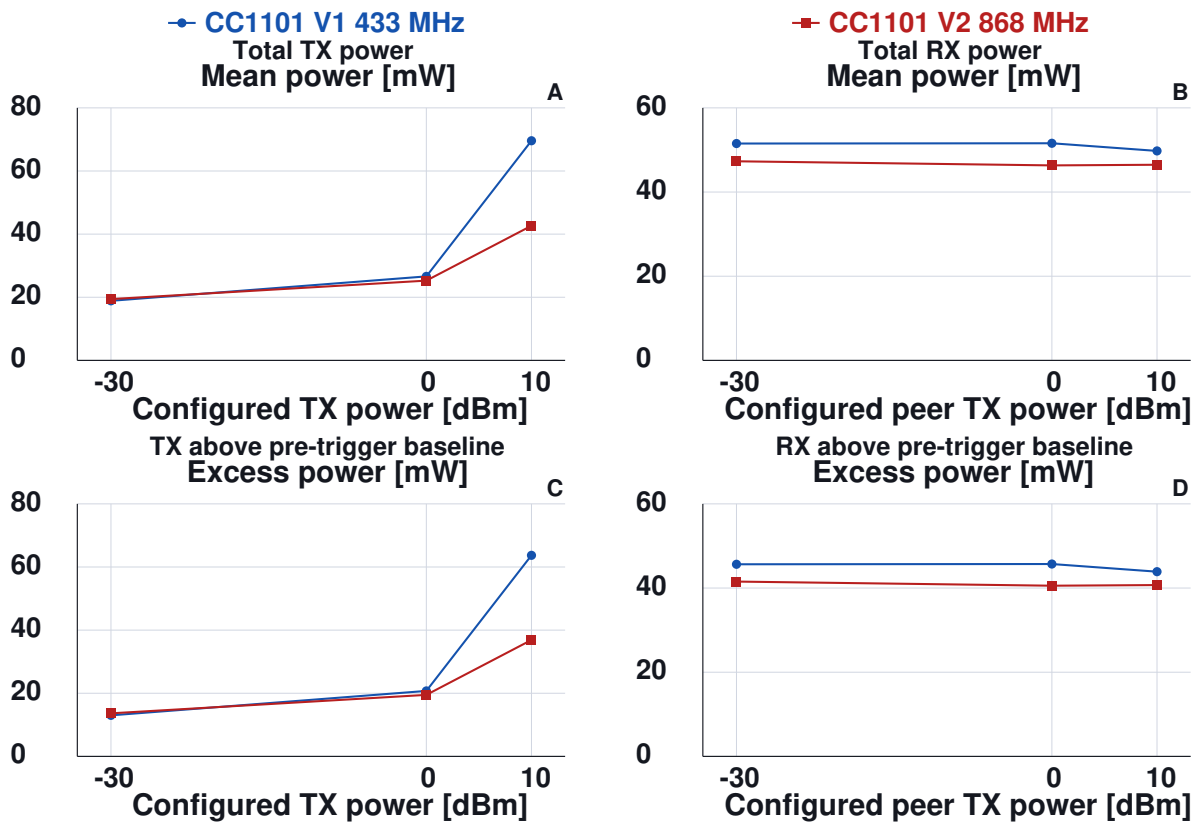


Figure 11: Controlled continuous-power comparison of the CC1101 V1 433-MHz and V2 868-MHz boards. Total and pre-trigger-baseline-subtracted values use the same 60-second, 32-byte, 38.4 kbps workload. These original sessions take the baseline with RX off. The configured power on RX is the peer transmitter setting.

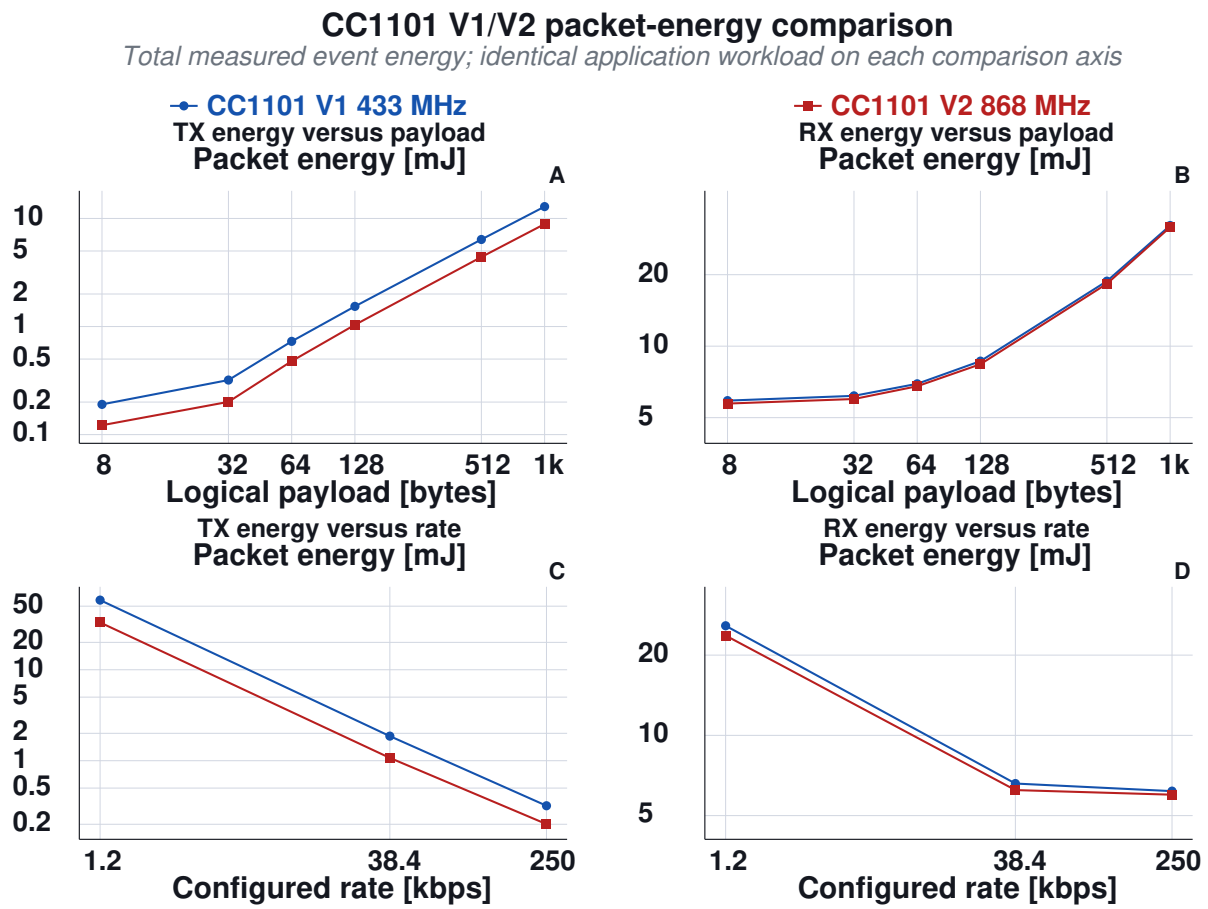


Figure 12: CC1101 packet-energy scaling on common axes. Payload panels use 250 kbps and +10 dBm; rate panels use a 32-byte payload and +10 dBm. Both axes are logarithmic.

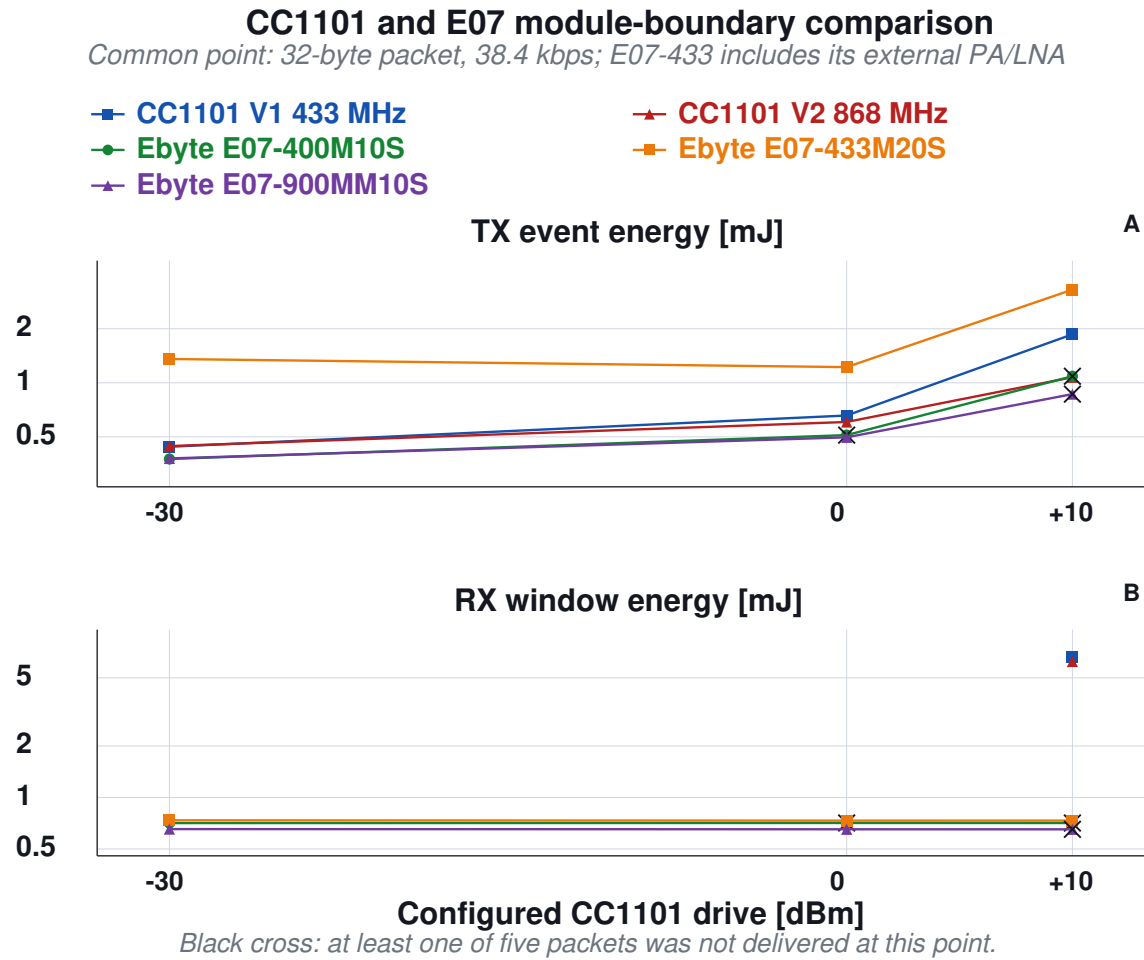


Figure 13: Module-boundary packet-energy comparison at a common 32-byte, 38.4 kbps GFSK/2-FSK workload. TX contains the full $-30/0/+10$ -dBm sweep for all five implementations. The legacy CC1101 RX campaigns contain only $+10$ -dBm peer points. A black cross marks an E07 point with incomplete delivery.

Ai-Thinker RA-08 (ASR6601) versus Ai-Thinker RA-09 (STM32WLE5): energy versus payload
Matched LoRa point: SF7/BW125/CR4/5 and +22 dBm

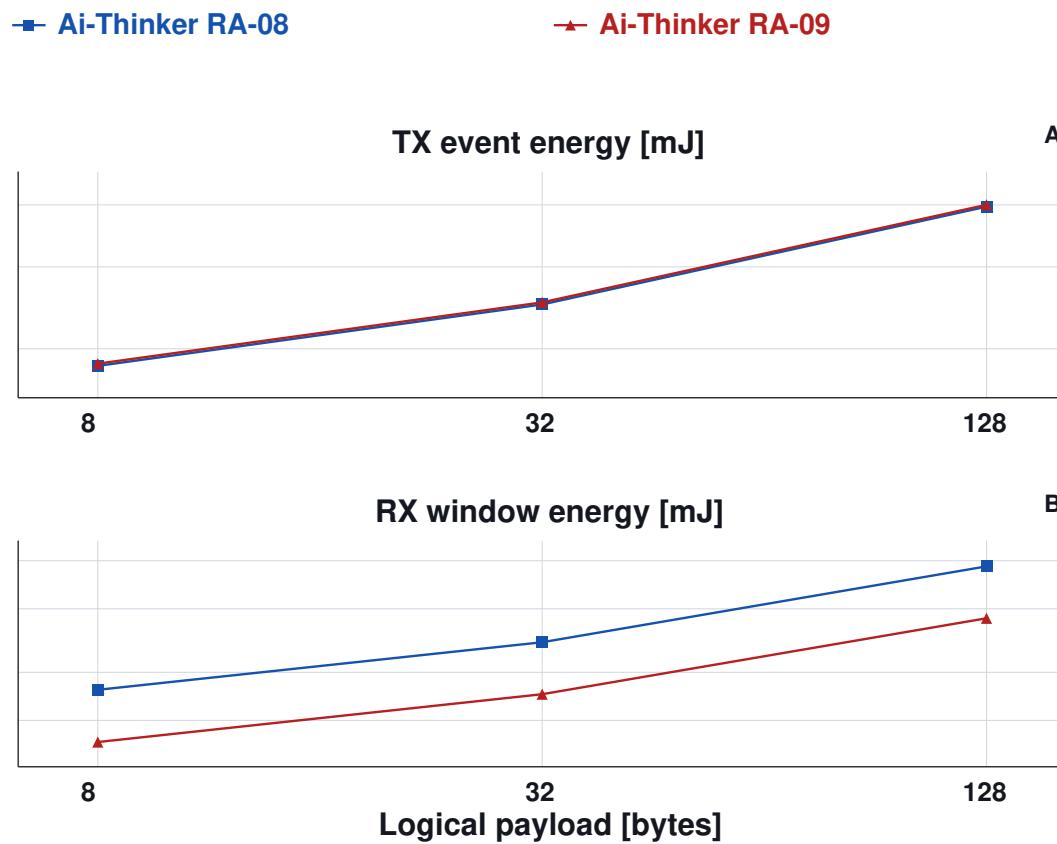


Figure 14: Matched Ai-Thinker RA-08 (ASR6601) and Ai-Thinker RA-09 (STM32WLE5) packet energy versus payload at SF7/BW125/CR4/5 and +22 dBm.

Ai-Thinker RA-08 (ASR6601) versus Ai-Thinker RA-09 (STM32WLE5): energy versus SF *Matched 32-byte LoRa point at BW125/CR4/5 and +22 dBm*

■ **Ai-Thinker RA-08**

▲ **Ai-Thinker RA-09**

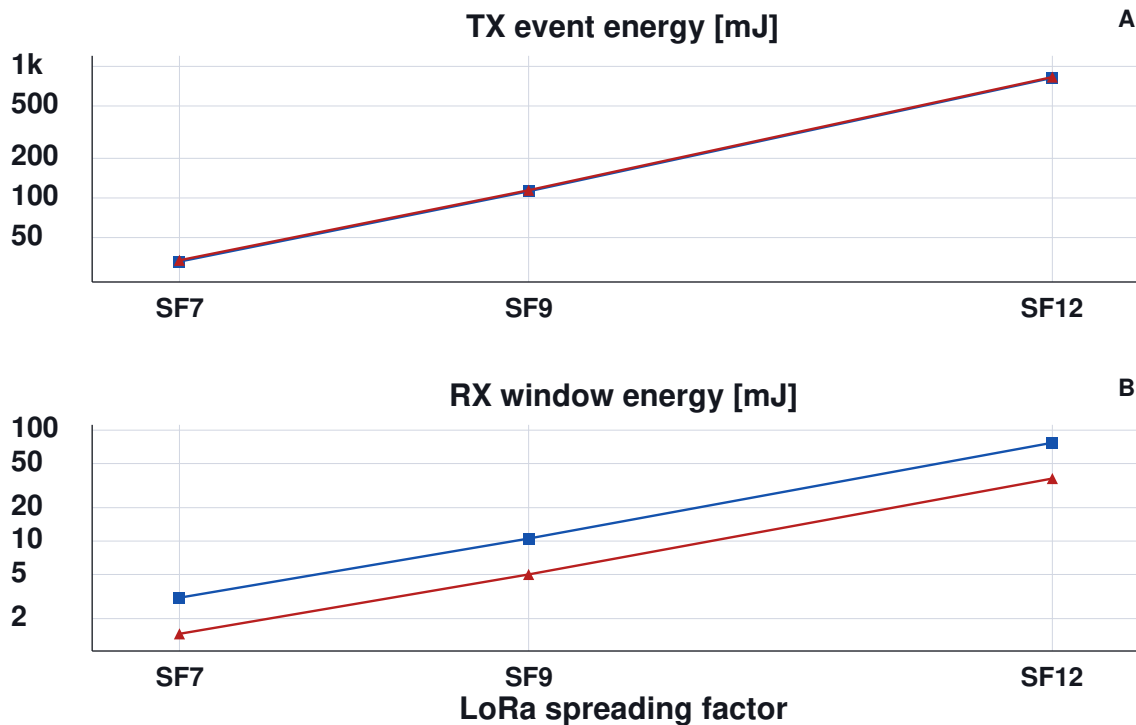


Figure 15: Matched Ai-Thinker RA-08 (ASR6601) and Ai-Thinker RA-09 (STM32WLE5) packet energy versus spreading factor for a 32-byte payload at BW125/CR4/5 and +22 dBm.

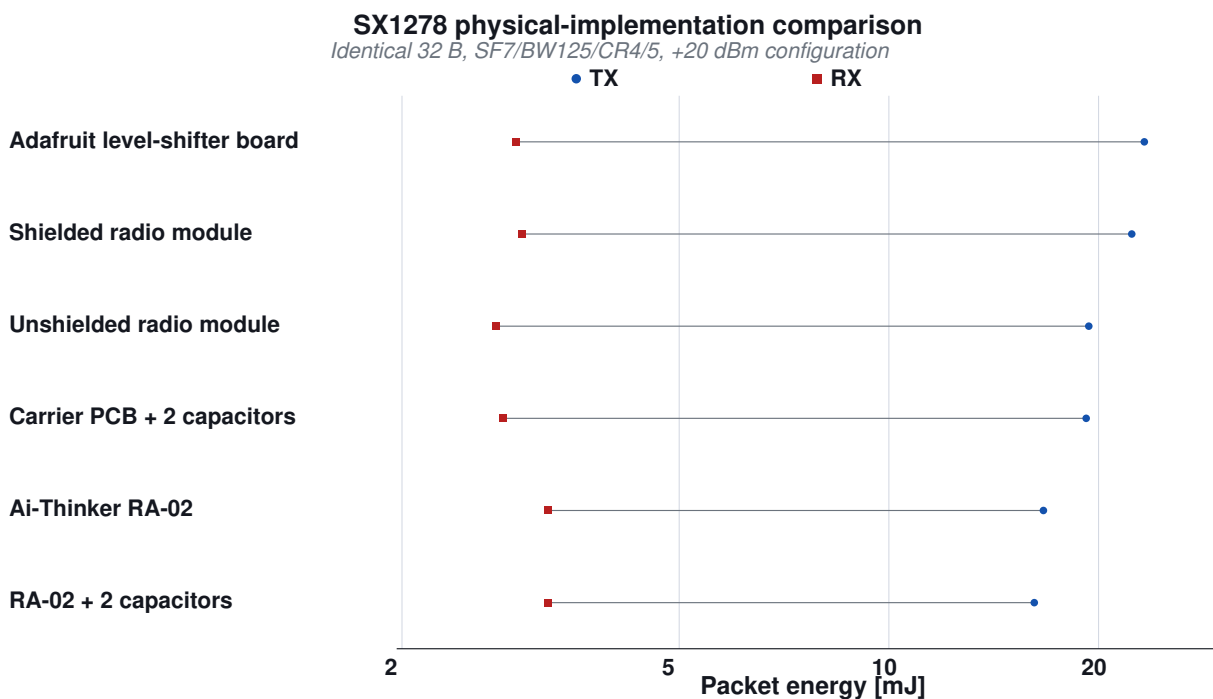


Figure 16: Controlled SX1278 comparison at 32 bytes, SF7/BW125/CR4/5, and +20 dBm. The same radio settings do not produce identical board-level energy.

These are module-boundary observations; the experiment does not isolate regulator quiescent current, RF matching loss, logic translation, and oscillator behavior sufficiently to assign a single cause.

Figure 17 extends the packet-only comparison to every available 60-second SF9/BW125 window. At +20 dBm, total TX power ranges from 196.94 mW for Ai-Thinker RA-02 with 2 Cap (SX1278) to 255.33 mW for Adafruit board with level shifter (SX1278), while total RX power occupies a narrower 37.75 – 41.37 mW interval. The Carrier PCB with 2 Cap (SX1278) RX point at -4 dBm is retained only as a power observation: zero received frames prevent it from supporting a useful-delivery conclusion.

Within this six-board view, the continuous TX reduction from adding two capacitors to RA-02 is 0.36 % at -4 dBm, 3.71 % at +10 dBm, and 3.31 % at +20 dBm. At +20 dBm, this corresponds to 196.94 mW versus 203.69 mW. RX total power differs by at most 0.3 %, so the added capacitors do not produce a meaningful receive-power advantage in this workload. The repeated few-percent TX effect at the two higher settings is measurable, but the experiment does not establish whether it comes from supply impedance, PA behavior, or RF matching.

The repository audit identifies four measured SX1276-based implementations: Ai-Thinker RA-01H (SX1276), XL1276-D01 (SX1276), Ebyte E32-868T20D (SX1276), and Ebyte E32-868T30D (SX1276 + PA). Figure 18 retains all four. At their canonical points, packet TX energy is 17.66 mJ, 21.35 mJ, 30.24 mJ, and 91.65 mJ for Ebyte E32-868T20D (SX1276), XL1276-D01 (SX1276), Ai-Thinker RA-01H (SX1276), and Ebyte E32-868T30D (SX1276 + PA), respectively. RX spans a much narrower 2.860 – 4.884 mJ. These values are useful module-level operating points, but the E32 transparent-UART framing, higher nominal rate, and the E32-868T30D external PA prevent a controlled four-way PHY comparison.

Within the matched direct-SPI subset, XL1276-D01 uses 21.35 mJ TX versus 30.24 mJ for RA-01H, a 29.4 % reduction, while RX is 2.934 mJ versus 2.860 mJ. In continuous operation at +20 dBm, XL1276-D01 averages 282.87 mW TX versus 367.36 mW for RA-01H; RX reverses the ordering at 40.78 mW versus 35.87 mW. These are same-chip, same-LoRa-setting observations, but not a pure PCB comparison because the RF bands, matching networks, and individual units differ.

The same audit found no additional measured SX1276 modules: RA-01 and the 433-MHz E32 units are SX1278 implementations, while RFM9x and inAir9 occur only as KiCad library footprints without campaigns. E28 and E280 both use SX1280 but expose direct-SPI LoRa and vendor transparent-UART PHYs, respectively. Their results remain valid system-level measurements, but their framing and control paths prevent attribution of a difference to the radio board alone.

The newer RA-01SH/SX1262 is notable: at its canonical +22 dBm point it uses 18.03 mJ TX and 1.185 mJ RX, versus 30.24 mJ and 2.860 mJ for the RA-01H/SX1276 at +20 dBm. Its matched continuous RX power is 14.76 mW versus 35.87 mW. The measured reduction is consistent with, but does not independently prove, the SX1262's newer low-power receiver and DC-DC architecture.

At 2 Mbps, the PA/LNA board uses 0.0935 mJ per TX event versus 0.0187 mJ for the plain module, almost exactly 5.0 times as much. In matched 1 Mbps continuous windows, TX power rises from 1.69 mW to 8.98 mW, and RX power from 41.8 mW to 73.3 mW. Across the drive sweep, the PA/LNA TX-power multiplier grows from 3.17 at -18 dBm to 5.30 at 0 dBm, while total RX power remains 1.68–1.75 times that of the plain module. The zero RX excess values in Figure 21 are baseline-clamped, not zero total consumption. The PA/LNA variant also delivered fewer frames in this bench setup. This does not imply that an external PA is intrinsically undesirable: its purpose is link budget. It shows that its energy cost must be justified with range or reliability measurements at the intended distance and supply quality.

The E32 20-dBm pair is the closest hardware match in the continuous campaign. At +20 dBm, Ebyte E32-433T20D (SX1278) and Ebyte E32-868T20D (SX1276) average 275.86 mW and 276.37 mW TX, while RX is 43.441 mW and 43.438 mW. Across all three power settings, the maximum TX difference is 0.52 % and the maximum RX difference is 0.11 %. The RX baseline-subtracted curve is clipped to zero by the campaign pipeline because the active-window mean falls below the pre-trigger listening mean; it must not be interpreted as a zero-power receiver. Total RX power is the valid comparison in this case.

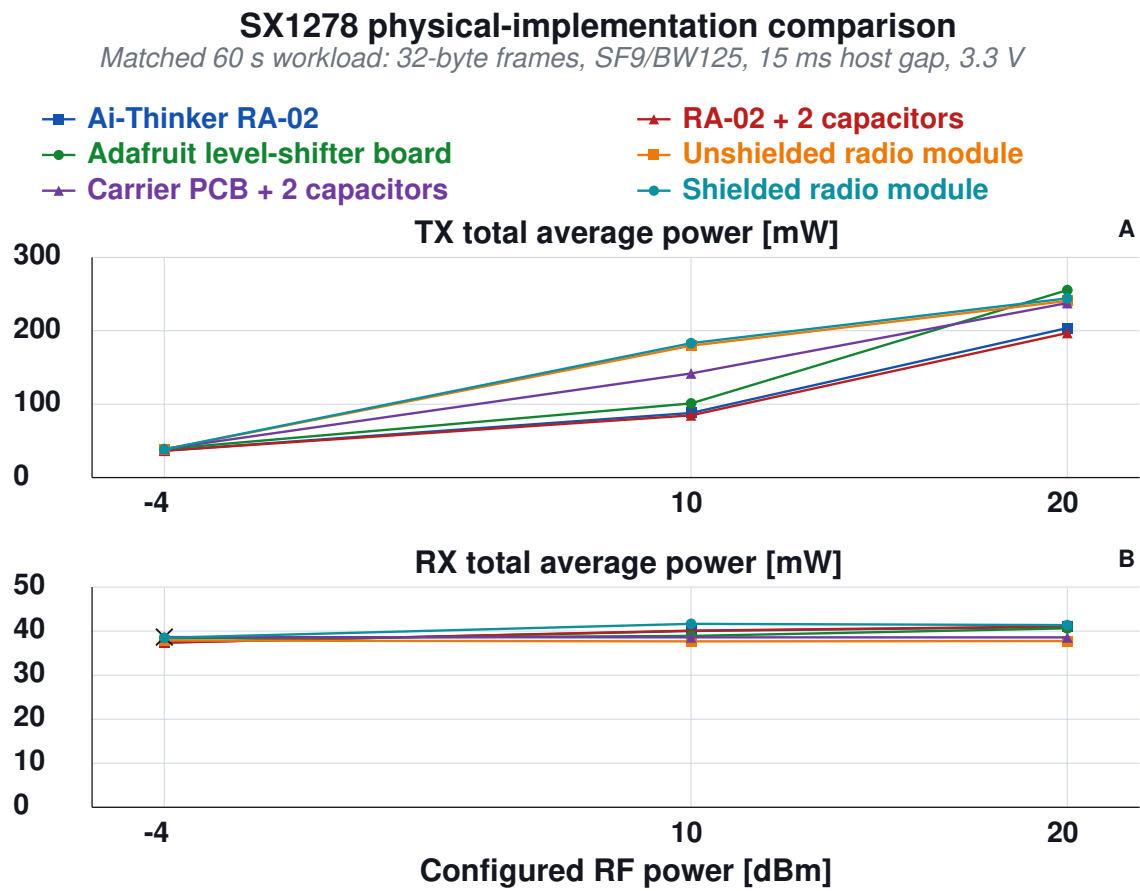


Figure 17: Matched continuous total-power comparison for all six SX1278 implementations from Figure 16. The TX and RX panels use independent vertical scales so that board-level differences remain visible.

SX1278 physical-implementation comparison (continued)

Matched 60 s workload: 32-byte frames, SF9/BW125, 15 ms host gap, 3.3 V

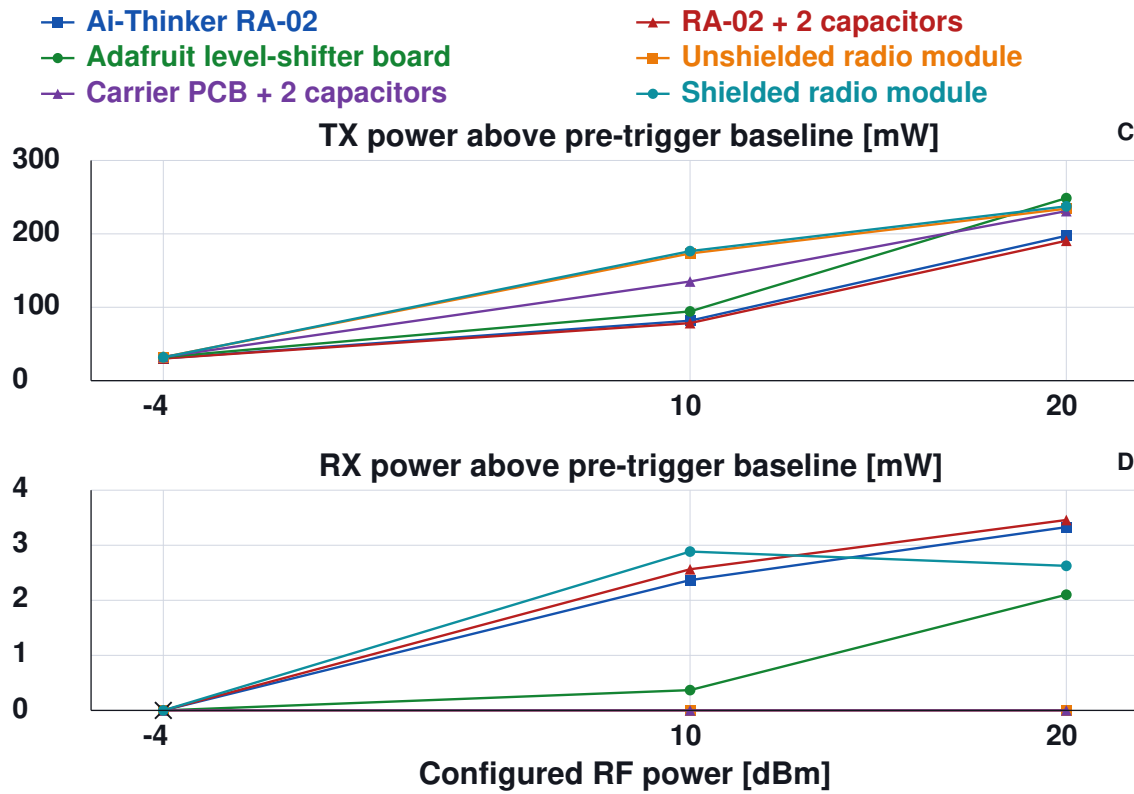


Figure 17: Matched continuous-power comparison for all six SX1278 implementations (continued): TX and RX power above each session's pre-trigger baseline. The RX reference is listening before traffic. A black cross marks the Carrier PCB with 2 Cap (SX1278) RX window at -4 dBm, whose power trace is valid but whose receiver-delivery telemetry is missing.

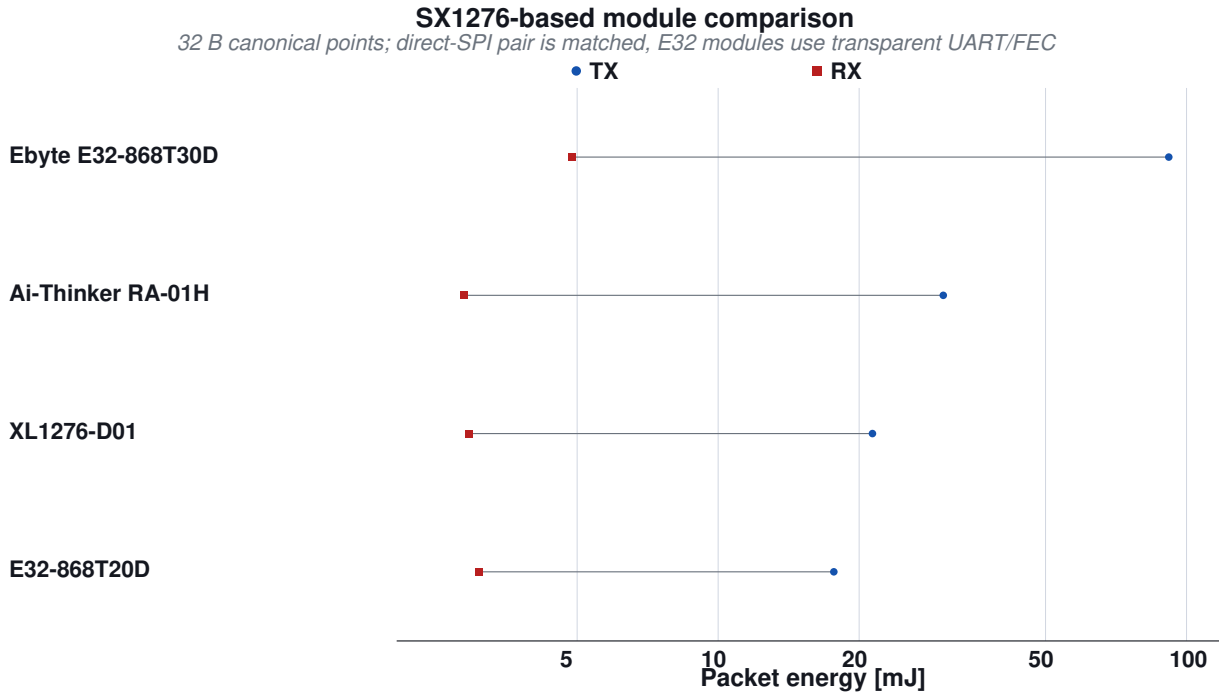


Figure 18: All four measured SX1276-based modules at their canonical 32-byte points. Ai-Thinker RA-01H (SX1276) and XL1276-D01 (SX1276) use direct SPI, SF7/BW125/CR4/5, and +20 dBm. Ebyte E32-868T20D (SX1276) and Ebyte E32-868T30D (SX1276 + PA) use transparent UART/FEC at 19.2 kbps and +20 dBm or +30 dBm, respectively. The latter includes an external PA, so the four points are a complete system-level inventory, not a single controlled operating condition.

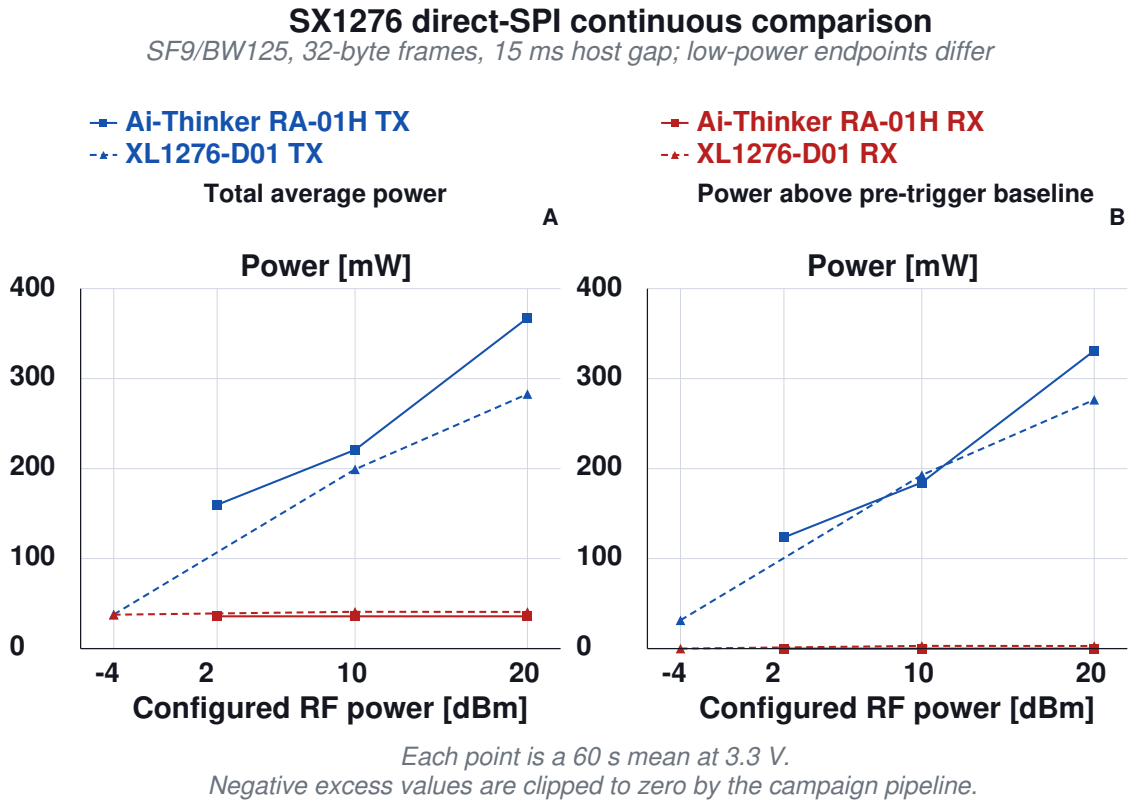


Figure 19: Controlled continuous-power comparison of the direct-SPI SX1276 subset, RA-01H and XL1276-D01, at SF9/BW125. Both include +10 and +20 dBm; the low endpoint is +2 dBm for RA-01H and -4 dBm for XL1276-D01.

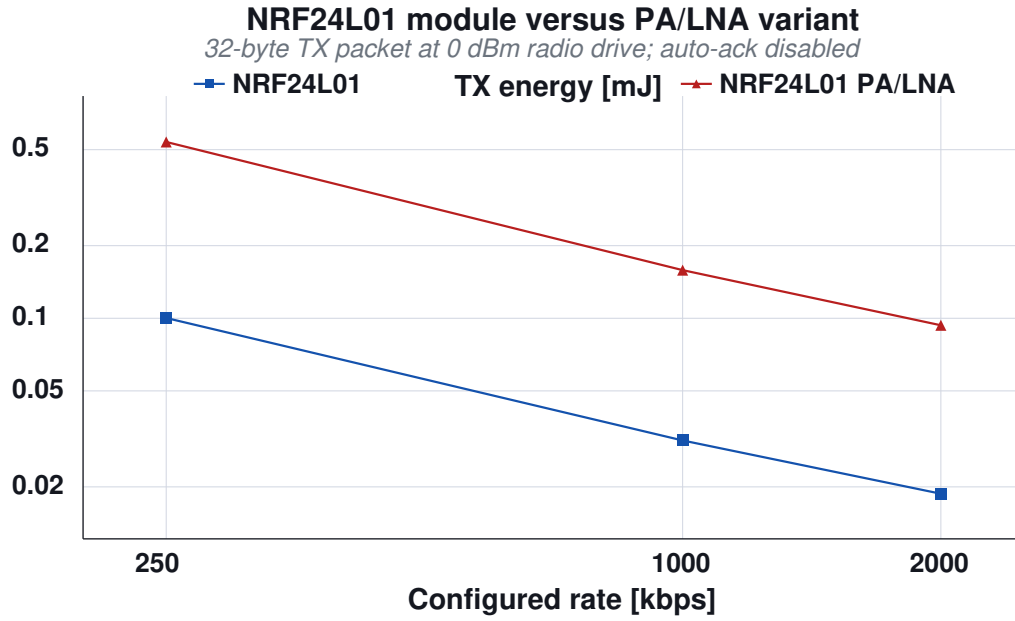


Figure 20: NRF24L01 (nRF24L01+) packet TX energy versus rate with and without the module-level PA/LNA. Both use the same 0 dBm radio-drive command.

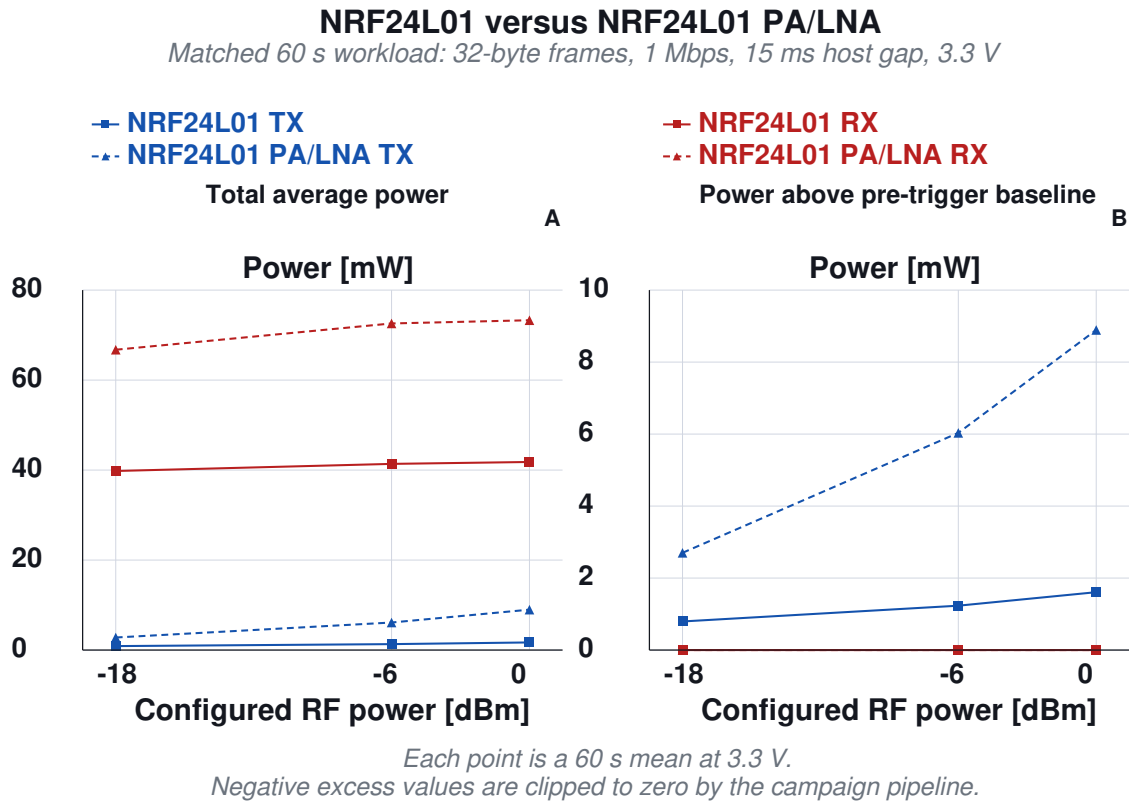


Figure 21: Matched NRF24L01 (nRF24L01+) continuous-power comparison at 1 Mbps. NRF24L01 PA/LNA (nRF24L01+) increases both TX and RX module-boundary power across all three configured radio-drive settings.

Ebyte E32-433T20D versus Ebyte E32-868T20D

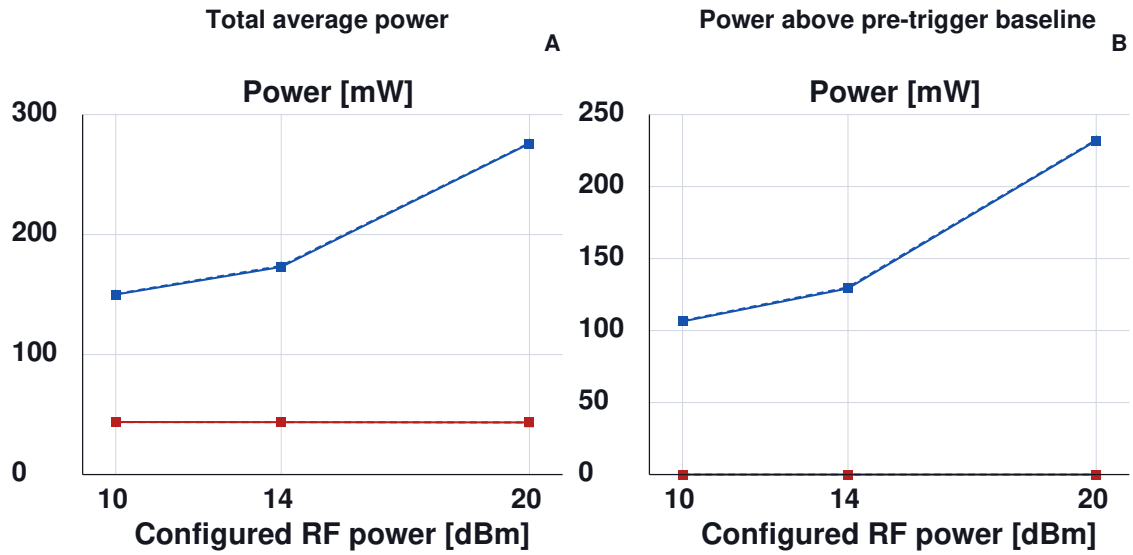
Matched 60 s workload: 58-byte frames, 4.8 kbps, 15 ms host gap, 3.3 V

—■ E32-433T20D TX

- - -■ E32-868T20D TX

—■ E32-433T20D RX

- - -■ E32-868T20D RX



Each point is a 60 s mean at 3.3 V.

Negative excess values are clipped to zero by the campaign pipeline.

Figure 22: Matched continuous-power comparison of Ebyte E32-433T20D (SX1278) and Ebyte E32-868T20D (SX1276). The nearly coincident curves show that these two measured modules behave alike under the common 4.8 kbps workload.

5 Discussion

5.1 Design implications

For short, frequent, nearby transfers, the measured high-rate NRF24L01 (nRF24L01+) and Ebyte E79-400DM2005S (CC1352P) GFSK profiles minimize packet airtime and absolute event energy at the common 32-byte workload. Payload curves show why application batching matters: a larger message costs more total energy but generally less energy per application byte. Ebyte E79-400DM2005S (CC1352P) is especially flexible because one firmware image can trade GFSK speed for SLR robustness, OOK behavior, or IEEE 802.15.4g framing. The bridge-context repetition shows that this PHY ordering is robust to the tested host-interface change, while the few-percent absolute offset should be treated as session variability until repeated bridge-swapping measurements are available. For long-range low-duty-cycle sensing, both Ai-Thinker RA-01SH (SX1262) and Ai-Thinker RA-09 (STM32WLE5) show strong measured RX-energy advantages over older LoRa implementations, while their TX behavior remains workload- and output-stage-dependent. High-power E32 UART modules simplify integration and provide substantial nominal output power, but their energy and buffering behavior make them a poor fit for frequent battery-powered traffic unless the link budget requires them.

The matched-pair curves sharpen several lessons. Ebyte E32-433T20D (SX1278) and Ebyte E32-868T20D (SX1276) remain nearly indistinguishable under the tested workload; Ebyte E07-400M10S (CC1101) and Ebyte E07-900MM10S (CC1101) likewise track closely at the fully delivered low-power point. In contrast, the Ebyte E07-433M20S (CC1101 + PA/LNA) external output stage changes TX energy substantially, and the two earlier CC1101 boards diverge at high TX drive despite sharing the same radio IC. A chipset name is therefore insufficient to predict module energy, but neither should every band or board revision be assumed to matter equally. Antenna size, propagation, legal channels, and interference remain consequential outside these energy traces.

5.2 What this study does not measure

No RF power meter, calibrated attenuator chain, conducted sensitivity setup, spectrum analyzer, or controlled-distance range course is part of this dataset. Configured dBm is a command or front-stage setting, not verified conducted output or EIRP. The study cannot compute joules per successfully delivered bit at equal range or equal fade margin. It also cannot rank coexistence, adjacent-channel rejection, antenna efficiency, spectral compliance, or immunity to real interference.

Temperature, unit-to-unit variation, battery internal resistance, and supply voltage were not swept. Five packet repetitions characterize repeatability at each point but are not a production tolerance study. The E79 bridge comparison was not collected as an alternating A/B/A/B experiment, so session drift is confounded with host-interface context. The five new entries do not yet have 60-second continuous campaigns. Some existing continuous campaigns intentionally retain no-RX scenarios and receiver re-arm limitations. Instrument accuracy, automatic range switching, event segmentation, modeled capture windows, and serial telemetry are additional uncertainty sources.

Finally, window energy is workload- and definition-dependent. Full-transaction battery budgeting requires an integration window that covers the intended transaction, including any relevant state transitions and gaps. The reported RX windows and marked historical CH9340C fragmented-TX windows must retain their stated limits. A silicon-only comparison should use conducted RF instrumentation and carefully separated idle, synthesizer, ramp, TX, RX, and host-processing intervals.

6 Reproducibility

The complete derivation is generated with only the Python standard library and the repository’s dependency-free PDF/PNG canvas:

```
python power_profiler/study/generate_study.py
pdflatex -interaction=nonstopmode radio_module_energy_study.tex
pdflatex -interaction=nonstopmode radio_module_energy_study.tex
```

Run the LaTeX commands from `power_profiler/study`. The generated `data/module_summary.csv` records the common 32-byte row selection, delivery scope, packet counts, selected modes, and derived metrics. The companion `data/payload_energy_summary.csv` records every TX/RX payload point used by Tables 2 and 3, including the secondary per-bit normalization. The new controlled packet exports are `data/e07_controlled_summary.csv`, `data/cc1101_family_summary.csv`, `data/e79_interface_summary.csv`, and `data/ra08_ra09_summary.csv`. The earlier CC1101 points remain available in `data/cc1101_controlled_summary.csv`. The E32, NRF24L01 (nRF24L01+), and Ai-Thinker RA-02 (SX1278) continuous sweeps are in `data/matched_continuous_power_`

`summary.csv`. Individual campaign validation notes and audit JSON files remain authoritative for accepted recoveries and known anomalies.

7 Conclusion

The measurements show that payload size, radio choice, PHY rate, module implementation, output stage, and receive-window current each materially affect energy. Across the common 32-byte points, fast GFSK devices minimize measured event energy, while high-output transparent LoRa modules incur the greatest cost. The payload matrices retain the measured event/window energies and expose campaign-specific frame and fragmentation boundaries; the marked historical CH9340C fragmented-TX values do not establish full-message energy. The Ebyte E79-400DM2005S (CC1352P) within-device study demonstrates a 38.5-fold 32-byte packet-energy swing from SLR2K5 to GFSK200 without changing hardware, and the non-fragmented bridge-context points preserve that PHY ordering. The E07 comparison isolates the cost of the PA/LNA implementation, while the matched Ai-Thinker RA-08 (ASR6601) and Ai-Thinker RA-09 (STM32WLE5) comparison finds similar TX energy but roughly half the Ai-Thinker RA-09 RX energy over equal LoRa windows. The SX1278 and NRF24L01 (nRF24L01+) sub-studies likewise show that board-level components can shift energy even when nominal radio settings are held constant.

The practical conclusion is not that one module wins every application. It is that module selection should be made with a workload-specific point containing payload, rate, output power, receive duty, required link margin, and delivered traffic behavior. This dataset supplies the energy half of that decision and identifies the next necessary experiment: conducted or distance-controlled link-budget and packet-error-rate testing under equal reliability targets.

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